

High-Dimensional Instrumental Variables Estimation of the Almost Ideal Demand System

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Abstract

The Almost Ideal Demand System by Deaton and Muellbauer (1980) is grounded in utility maximization and is flexible, aggregates exactly over heterogeneous consumers, and delivers interpretable own and cross-price elasticities. However, these properties come at a cost: the number of cross-price parameters grows quadratically in the number of goods, and since prices are typically endogenous, each requires a separate instrument. Modern scanner data routinely contains hundreds of goods and AIDS has consequently been confined to small systems or displaced by alternative specifications that sacrifice the interpretability and aggregation properties. I argue that AIDS was sidelined not because its economic structure is ill-suited to high-dimensional settings, but because the estimation tools required to apply it at scale did not exist. I adapt the doubly-high-dimensional debiased estimator of Gold et al. (2020) to the AIDS share equation and show via simulation that this estimator delivers stable estimates of cross-price elasticities in high-dimensional settings where classical methods break down.

1 Introduction

The Almost Ideal Demand System of Deaton and Muellbauer (1980) is a specification for empirical demand analysis grounded in utility maximization. The system is derived from the flexible PIGLOG cost function, and its budget-share equations can provide a first-order local approximation to any demand system consistent with consumer theory. The demand system aggregates exactly over heterogeneous consumers and delivers interpretable own and cross-price elasticities. These properties make AIDS a desirable tool for consumer demand analysis across applied microeconomics, agricultural economics, and public finance.

The benefit of flexibility, however, carries a cost. In a system with n goods, the matrix of cross-price parameters contains n^2 entries. Utility maximization provides restrictions of adding-up, homogeneity, and Slutsky symmetry, which reduce the free-parameter count but cannot eliminate its quadratic growth. A second difficulty is that prices are typically endogenous in applied demand settings since market prices and consumer demand are jointly determined in the presence of unobserved shocks. The issue is that identifying each of n endogenous prices requires at least n instruments, so the instrument count must also scale with the number of goods. Moreover, modern scanner data contains hundreds of goods and potentially thousands of candidate cost-shifters and rival-market prices that could serve as instruments, making classical AIDS estimation numerically infeasible.

The leading response to this dimensionality issue has been multi-stage budgeting as per Gorman (1959), in which similar products are grouped together and AIDS is applied within each group, reducing the number of parameters. A weakness of this approach, however, is that it restricts the substitution pattern between groups.¹ The accuracy of this approach depends on the grouping, which can be ambiguous *ex ante*, and incorrect groupings yield biased cross-price elasticities. Given these limitations, the alternative at the scale of modern data has been to abandon AIDS altogether, despite it remaining a flexible functional form for interpretable own and cross-price elasticities in continuous demand systems.²

¹Hausman et al. (1994) segment beer into premium, light, and standard; a price change in a premium beer, for example, then affects light demand indirectly only through the group-level price index.

²Large-scale demand analysis is now dominated by the characteristic-space paradigm of Berry et al. (1993), Berry (1994), and Nevo (2001), in which products are represented by a low-dimensional vector

I adapt recent advances from modern machine learning methods to argue that AIDS was sidelined not because it is the wrong economic model, but because the estimation tools required to apply it at scale did not exist. The central methodological tool is the doubly-high-dimensional debiased one-step estimator of Gold et al. (2020), which extends the many-instruments post-Lasso-IV program of Belloni et al. (2012) to the regime in which both the number of endogenous regressors and the number of instruments can be large relative to the sample size. In the AIDS context, the endogenous regressor vector is the n -dimensional log-price vector and the candidate instrument set may contain hundreds of cost-shifters or lagged prices of related goods. The estimator relies on approximate sparsity, meaning that demand for a product depends on only a small number of close substitutes rather than every other good.³ Li et al. (2018) provides a market-structure argument for this, applying high-dimensional methods to demand estimation and arguing that own-demand depends on a small set of rivals because firms have limited awareness of distant competitors and the cost of tracking them makes responding to their prices suboptimal. A parallel sparsity assumption applies to the first stage: each endogenous price is well-predicted by a small subset of the candidate instruments.⁴

I add two practical modifications to the debiased-IV backbone. First, I cross-fit the first stage following Chernozhukov et al. (2018), partitioning the sample so each observation’s first-stage prediction is built from held-out data. The construction is designed to remove the in-sample dependence between the first-stage fit and the structural residuals, which can distort finite-sample coverage in principle. Gold et al. (2020) do not require cross-fitting for asymptotic validity. Second, I use heteroskedasticity-robust sandwich standard errors, since Gold et al. (2020) maintain homoskedasticity in their main theorem while budget-share errors are heteroskedastic by construction. Caner and Kock (2018) establish validity of a related debiased IV framework under conditional heteroskedasticity in the structural error, which supports this choice.

To assess performance, I conduct Monte Carlo experiments varying the number of

of observable attributes and demand is derived from discrete-choice foundations. This approach has been enormously productive, and this paper is not an argument against it.

³Approximate sparsity is sufficient: the framework assumes most cross-price coefficients are small in magnitude rather than exactly zero, and does not require prior knowledge of which coefficients those are.

⁴This is a standard assumption in the high-dimensional IV literature, following Belloni et al. (2012) and Gold et al. (2020).

goods and the instrument count, while holding the sample size T fixed. OLS is inconsistent under endogeneity and becomes rank-deficient as the number of goods approaches the sample size. As the instrument count grows, classical 2SLS becomes rank-deficient, and Post-Lasso IV handles many instruments but not many endogenous regressors simultaneously. The debiased HD-IV estimator extends through the high-dimensional regime. Comparators spike as n approaches T while the framework’s ridge and nodewise variants maintain the RMSE level held across the inner grid. Coverage of the own-price coefficient is somewhat below the nominal level and narrows toward it as the sample grows. Coverage of the Engel-curve slope is poorer in finite samples.⁵ The debiased framework does not solve weak identification, which remains a challenge for all estimators. The gains here are in numerical stability across high-dimensional regimes.

To my knowledge, no prior work has applied debiased regularized IV methods to AIDS or documented their finite-sample behavior as the number of goods and instruments grows. The methodology draws entirely on existing results; the contribution lies in assembling them for the AIDS setting and showing that AIDS is estimable at high dimensions. The simulation results confirm that the model remains well-behaved at scales the classical literature treated as infeasible, if modern machine learning tools are applied correctly.

The rest of the paper is organized as follows. Section 2 reviews the relevant literature. In Section 3, I introduce the AIDS model and formalize the dimensionality problem. In Section 4, I develop the econometric framework, presenting the two-stage Lasso and the debiased one-step update. In Section 5, I describe the Monte Carlo design and present the simulation results. Section 6 concludes.

2 Literature

The paper draws on four strands of literature that have developed largely independently.

Classical AIDS estimation with endogenous prices has a developed toolkit, most prominently the iterated linear least squares procedure of Blundell and Robin (1999) and its

⁵This is a known limitation of debiased Lasso for small-magnitude coefficients rather than a consequence of dimensionality. The Engel slope is a single small coefficient, while the high-dimensional challenge in AIDS comes from the cross-price block, where coverage holds up better.

Stata implementation in Lecocq and Robin (2015).⁶ However, this toolkit was designed for moderate dimensions and does not address high-dimensional regimes or instrument-selection problems. Blundell and Robin (1999) exploit the conditional linearity of AIDS in its structural parameters and proves that the procedure is consistent and asymptotically normal under standard conditions. Representative applications include Deaton (1988), Nelson (1997), Chouinard et al. (2007), and Fajgelbaum and Khandelwal (2016).⁷ This literature accomplishes consistent IV-based estimation when the number of goods is small relative to the sample size but does not address the regime where the parameter count and instrument count are both comparable to or exceed the sample size, which is the focus of this paper.

The most direct precedent for applying regularization to AIDS is Bae (2018), which applies elastic-net penalization in a four-good Monte Carlo replicating the design of Buse (1994) and shows that regularization improves finite-sample performance under multicollinearity and sparsity. It documents the value of shrinkage in a demand context without addressing endogeneity. Two broader approaches reduce dimension through structure rather than sparsity. Fally (2022) characterizes generalized separability, in which cross-price effects operate through a small number of scalar aggregators, yielding low-rank substitution matrices consistent with utility maximization. Hausman (1996) and Hausman et al. (1994) reduce dimension through multi-stage budgeting by imposing structure on product groups *ex ante*. Both approaches are useful when their structural assumptions are credible, but those assumptions (separable aggregators or the right product groupings) are rarely known *a priori* and misspecification yields biased elasticities. The sparsity-based approach requires no prior knowledge of which cross-price effects are zero, only that most are small in magnitude, a condition that Li et al. (2018) justifies through a market-structure argument.

The methodologically closest paper is Chernozhukov et al. (2019), which articulates the many-prices problem in demand estimation and uses a Lasso first stage together with a

⁶Direct 3SLS on the full nonlinear AIDS system is an alternative at small n ; it accounts for cross-equation error correlation induced by adding-up but is computationally demanding and does not extend to high dimensions.

⁷Deaton (1988) estimates quality-adjusted unit-value demand with spatial price variation; Nelson (1997) applies AIDS to US alcohol demand with tax-rate instruments; Chouinard et al. (2007) estimates a dairy fat-tax counterfactual with cost-shifter instruments; Fajgelbaum and Khandelwal (2016) estimates a non-homothetic AIDS at the industry level across many countries.

double/debiased ML construction of the Riesz representer to estimate welfare functionals on scanner data.⁸ The shared premise is that approximate sparsity of cross-price effects is credible and that regularization is the natural machinery for exploiting it. Chernozhukov et al. (2019) target welfare functionals and do not impose AIDS integrability restrictions; this paper targets individual structural parameters β_i and γ_{ij} of an AIDS share equation directly. Adjacent contributions include Chernozhukov et al. (2022) and Farrell et al. (2021); neither addresses AIDS-specific parameters or restrictions, but both provide foundations that could be layered on top of the estimator developed here.⁹

The econometric backbone draws on three developments in high-dimensional IV inference. Belloni et al. (2012) and Belloni et al. (2014) solve the many-instruments problem (fixed endogenous regressors and a large instrument set) with Lasso-based first-stage selection. They prove $\sqrt{T}$ -consistency and asymptotic normality under approximate sparsity. Zhang and Zhang (2014), Van de Geer et al. (2014), and Javanmard and Montanari (2014) construct valid confidence intervals for individual coefficients in high-dimensional regression without endogeneity via a one-step Newton correction using an approximate inverse of the Gram matrix.¹⁰ Gold et al. (2020) combine both into the doubly-high-dimensional IV setting with many endogenous regressors and many instruments simultaneously and establish componentwise asymptotic normality under sparsity of both stages; this is the direct backbone of the estimator I develop. Several adjacent papers (Lin et al. (2015) and Zhu (2018)) propose estimation methods when both stages are high-dimensional but do not develop valid asymptotic inference; Neykov et al. (2018) provide an inferential method via Dantzig selector but require the instrument count to equal the endogenous regressor count.¹¹ Overlaying all of this is the double/debiased ML framework of Chernozhukov et al. (2018), which unifies Neyman-orthogonal moment construction with cross-fitting for

⁸Lasso, proposed by Tibshirani (1996), penalizes coefficient magnitudes to produce sparse estimates; see Bühlmann and Van De Geer (2011) for a comprehensive treatment.

⁹The scanner-data era, the prevalence of hundreds of barcodes per category, and the policy relevance of cross-price effects in tax, antitrust, and merger analysis motivate high-dimensional demand estimation; see Dubé et al. (2010), Handbury and Weinstein (2015), and Redding and Weinstein (2020).

¹⁰The approximate inverse is constructed via the nodewise-Lasso of Meinshausen and Bühlmann (2006) or the CLIME program of Cai et al. (2011).

¹¹Caner and Kock (2018) develop a parallel framework that allows conditional heteroskedasticity without sub-Gaussianity via a de-sparsified ℓ_1 -penalized two-stage GMM estimator. The impossibility results of Leeb and Pötscher (2005, 2008) motivate the debiased approach over post-selection inference. Running OLS on Lasso-selected variables does not yield uniformly valid confidence intervals, while the debiased estimator does.

$\sqrt{T}$ inference. I apply cross-fitting from this framework as a finite-sample refinement beyond what Gold et al. (2020) require asymptotically. This high-dimensional IV literature develops the tools for the many-regressor, many-instrument regime that applied demand estimation operates in, but has not been brought to AIDS with its structural restrictions. The present paper fills this gap.

3 The Almost Ideal Demand System

3.1 Model and restrictions

Deaton and Muellbauer (1980) derive AIDS from the PIGLOG cost function

$$\log c(u, p) = (1 - u) \log a(p) + u \log b(p), \quad (1)$$

where $u \in [0, 1]$ indexes utility between subsistence and bliss. For a system of n_{goods} goods indexed by $i, j, k = 1, \dots, n_{\text{goods}}$ with price vector $p = (p_1, \dots, p_{n_{\text{goods}}})$, $\log a(p)$ and $\log b(p)$ are specified as

$$\log a(p) = \alpha_0 + \sum_{k=1}^{n_{\text{goods}}} \alpha_k \log p_k + \frac{1}{2} \sum_{k=1}^{n_{\text{goods}}} \sum_{j=1}^{n_{\text{goods}}} \gamma_{kj}^* \log p_k \log p_j, \quad (2)$$

$$\log b(p) = \log a(p) + \beta_0 \prod_{k=1}^{n_{\text{goods}}} p_k^{\beta_k}. \quad (3)$$

The PIGLOG class is attractive because, by the aggregation theorems of Muellbauer (1975, 1976), market demands can be written as if generated by a rational representative consumer without requiring parallel linear Engel curves.

Applying Shephard's Lemma, $\partial \log c(u, p) / \partial \log p_i = w_i$, and substituting the indirect utility function $u = v(p, x)$ obtained by inverting $c(u, p) = x$, delivers the AIDS budget-share equation

$$w_{it} = \alpha_i + \sum_{j=1}^{n_{\text{goods}}} \gamma_{ij} \log p_{jt} + \beta_i \log \frac{x_t}{P_t}, \quad (4)$$

where w_{it} is the budget share of good i at observation t , indexed by $t = 1, \dots, T$ across the sample (typically time periods in aggregate demand data), x_t is total expenditure,

$\gamma_{ij} = \frac{1}{2}(\gamma_{ij}^* + \gamma_{ji}^*)$, and P_t is the exact price index

$$\log P_t = \alpha_0 + \sum_{k=1}^{n_{\text{goods}}} \alpha_k \log p_{kt} + \frac{1}{2} \sum_{j=1}^{n_{\text{goods}}} \sum_{k=1}^{n_{\text{goods}}} \gamma_{kj} \log p_{kt} \log p_{jt}. \quad (5)$$

Utility maximization imposes three sets of linear restrictions on the parameters (α, β, Γ) :

$$\text{Adding-up: } \sum_{i=1}^{n_{\text{goods}}} \alpha_i = 1, \quad \sum_{i=1}^{n_{\text{goods}}} \gamma_{ij} = 0 \quad \forall j, \quad \sum_{i=1}^{n_{\text{goods}}} \beta_i = 0, \quad (6)$$

$$\text{Homogeneity: } \sum_{j=1}^{n_{\text{goods}}} \gamma_{ij} = 0 \quad \forall i, \quad (7)$$

$$\text{Slutsky symmetry: } \gamma_{ij} = \gamma_{ji} \quad \forall i \neq j. \quad (8)$$

These restrictions are testable and, when imposed, reduce the free cross-price parameter count from n_{goods}^2 to $\frac{n_{\text{goods}}(n_{\text{goods}}-1)}{2}$.¹² The α and β vectors contribute $n_{\text{goods}} - 1$ free parameters each after adding-up. The total free-parameter count across the system is therefore

$$\frac{n_{\text{goods}}(n_{\text{goods}}-1)}{2} + 2(n_{\text{goods}}-1) = \frac{n_{\text{goods}}^2 + 3n_{\text{goods}} - 4}{2}. \quad (9)$$

This grows quadratically in n_{goods} . With $n_{\text{goods}} = 50$ goods the cross-price block alone contains 1,225 free parameters; with $n_{\text{goods}} = 200$ goods it contains 19,900.

3.2 The linear approximation

Since the exact price index P_t in (5) is nonlinear in parameters, direct estimation of (4) requires nonlinear least squares or maximum likelihood. Deaton and Muellbauer (1980) show that when prices are closely collinear, P_t can be adequately approximated by the Stone price index $P_t^* = \prod_j p_{jt}^{w_{jt}}$, yielding the linear-approximate AIDS (LA-AIDS) share equation

$$w_{it} = \alpha_i + \sum_{j=1}^{n_{\text{goods}}} \gamma_{ij} \log p_{jt} + \beta_i \log \frac{x_t}{P_t^*} + u_{it}, \quad (10)$$

¹²With Slutsky symmetry, the $n_{\text{goods}} \times n_{\text{goods}}$ matrix Γ has $\frac{n_{\text{goods}}(n_{\text{goods}}+1)}{2}$ unique entries. Homogeneity imposes n_{goods} constraints (one per row). Under symmetry, satisfying homogeneity for all rows also satisfies the adding-up constraint on columns, so the restrictions are not independent; the net free count is $\frac{n_{\text{goods}}(n_{\text{goods}}+1)}{2} - n_{\text{goods}} = \frac{n_{\text{goods}}(n_{\text{goods}}-1)}{2}$.

where u_{it} captures econometric error and the approximation error from replacing the exact price index P_t with the Stone approximation P_t^* , and satisfies $\sum_i u_{it} = 0$ by adding-up. This paper works with the LA-AIDS specification (10).¹³ Equation (10) is linear in $(\alpha_i, \beta_i, \gamma_{i1}, \dots, \gamma_{i, n_{\text{goods}}})$, so standard linear estimators apply once the regressors are constructed. The restrictions (6)–(8) translate into linear constraints on the stacked parameter vector. The simulations in Section 5 work with a single share equation and do not impose these constraints during estimation; they hold at the data-generating process by construction, and joint estimation that imposes them is left for future work.

3.3 The dimensionality problem

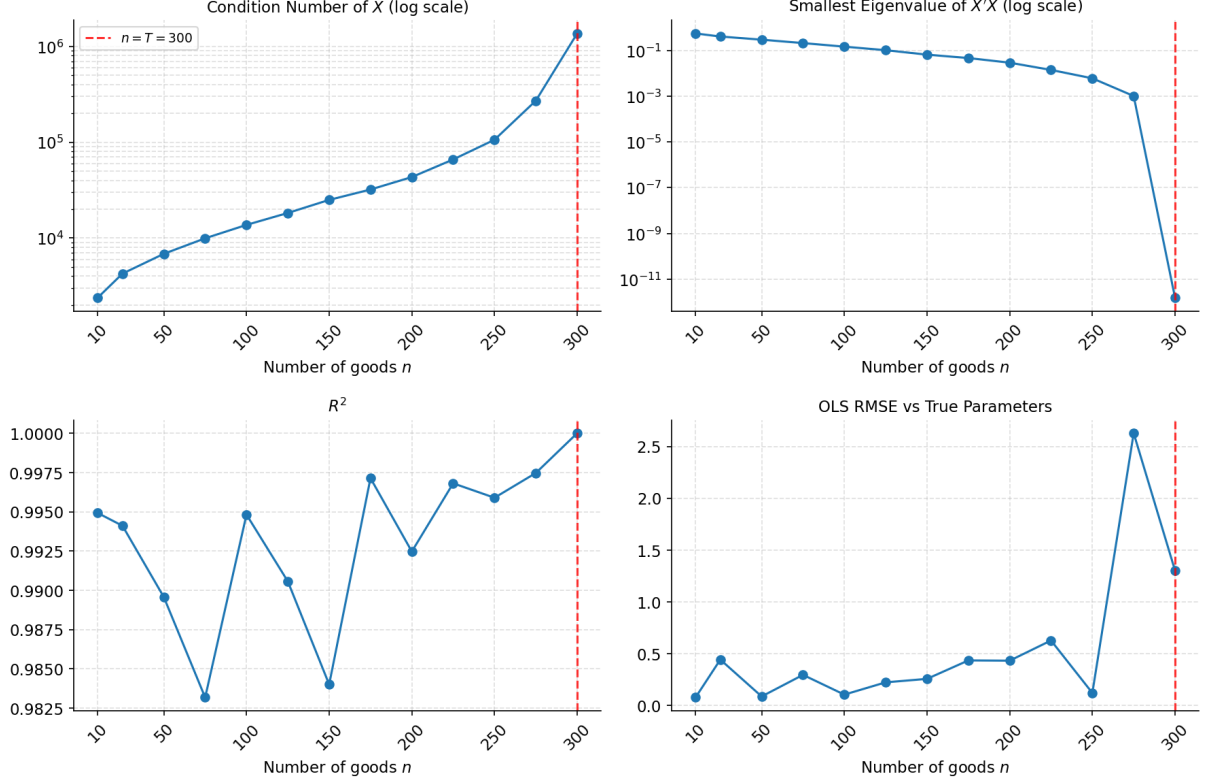
Two distinct problems arise as the number of goods grows. The first is *ill-conditioning*: even when $n_{\text{goods}} < T$ and OLS is technically feasible, highly correlated prices make the design matrix near-singular, the condition number grows sharply, and parameter estimates become sensitive to small perturbations in the data. The second is *rank deficiency*: once $n_{\text{goods}} + 2 > T$, the design matrix has more columns than rows and OLS is undefined. Ill-conditioning produces unreliable estimates well before the rank-deficient boundary, while rank deficiency is a hard wall at $n_{\text{goods}} + 2 = T$.

I illustrate both with a toy simulation. I simulate an LA-AIDS system with a common price factor C_t driving $\log p_{jt} = C_t + \varepsilon_{jt}$ ($\rho = 0.95$, producing the high collinearity typical of real-world price data), draw parameters under the AIDS restrictions (6)–(8), construct shares from (10), and estimate the share equation for good 1 by OLS. Sample size is fixed at $T = 300$ to match the main Monte Carlo simulation, and the number of goods n_{goods} varies from 10 to 300. For each n_{goods} , I generate one random replication of the simulation and record four diagnostics on the resulting design matrix $X \in \mathbb{R}^{T \times (n_{\text{goods}} + 2)}$ (with columns for the constant, log real expenditure, and the n_{goods} log-prices): the condition number of X , the smallest eigenvalue of $X'X$, the RMSE between the OLS estimates and the true parameters, and the R^2 of the regression. The first two capture the conditioning of the linear system, the third measures parameter recovery, and the fourth captures regression

¹³A literature documents the costs of this approximation. Pashardes (1993) shows the Stone-index substitution biases cross-price parameters, Buse (1994) establishes inconsistency of SUR on LA-AIDS, Alston et al. (1994) quantifies the resulting elasticity errors, and Moschini (1995) shows the Stone index is not unit-invariant. These issues are orthogonal to the dimensionality problem studied here.

fit. Variability across replications is examined separately in the Monte Carlo diagnostics reported in Appendix A.

Figure 1. Dimensionality problem under correlated prices



Notes: Simulation with $\rho = 0.95$ and $T = 300$. Clockwise from top-left: condition number of X (log scale), smallest eigenvalue of $X'X$ (log scale), OLS RMSE between estimated and true parameters, R^2 of the regression. Dashed line marks $n_{\text{goods}} = T$.

In this simulation with high price collinearity, I find that as n_{goods} grows toward T : (1) the condition number of X grows by several orders of magnitude; (2) the smallest eigenvalue of $X'X$ tends to zero; (3) the RMSE between OLS estimates and the true parameters rises sharply near the rank-deficient boundary; and (4) the R^2 of the regression stays near unity throughout, including at the rank-deficient boundary itself.

Patterns (1) and (2) reflect that as n_{goods} grows toward T , the design matrix accumulates more correlated columns, making the linear system progressively harder to invert. Pattern (3) demonstrates that OLS no longer recovers the true parameters reliably. Pattern (4) is worth highlighting. High explanatory fit is no protection against parameter recovery failure.¹⁴ A researcher who reports only R^2 as a diagnostic can completely miss

¹⁴When prices are highly correlated, many different coefficient vectors produce nearly identical fitted

the dimensionality problem. The RMSE pattern is also informative since it is not monotonic in n_{goods} . Individual draws can produce good or bad estimates depending on the realized correlation structure, which is a symptom of ill-conditioning as small changes in the data can produce large changes in the OLS solution.

To isolate the role of price collinearity, I run two contrasting designs, reported in Appendix A. With independent AR(1) prices, the conditioning deteriorates modestly compared to the correlated case, and OLS RMSE stays low until the rank-deficient boundary at $n_{\text{goods}} + 2 = T$. With perfectly orthogonal regressors, although the conditioning is unaffected by dimension, RMSE still rises sharply at the rank-deficient boundary $n_{\text{goods}} + 2 = T$. This shows that high dimensionality alone is enough to cause problems. Correlation among prices then makes the problem considerably worse, pushing the deterioration well before the boundary, as documented in Figure 1. Two further diagnostics in the appendix confirm this. The first, perturbation sensitivity, measures the change in the OLS coefficient vector $\hat{\beta}$ when small noise is added to X . The second is Monte Carlo confidence bands for the own-price coefficient γ_{11} across 100 replications. Under correlated prices, both quantities explode as n_{goods} approaches T . The takeaway is that as n_{goods} approaches T in correlated data, precision collapses and the ability to recover the true parameters degrades, even though the regression can continue to fit the data well in-sample. Real market prices are correlated by construction, so applied AIDS estimation faces this collapse.

An additional layer of difficulty arises when prices are endogenous, as they typically are in applied demand settings since market prices and unobserved demand shocks are jointly determined. Each of the n_{goods} endogenous log-prices requires at least one instrument, so the instrument count p_z must also scale with n_{goods} . Modern empirical demand applications compound this problem. They routinely yield hundreds of candidate cost-shifters, lagged prices, and rival-market prices that could serve as instruments, so p_z is typically large by virtue of data availability. When $p_z > T$, the first-stage instrument matrix is rank-deficient and classical two-stage least squares is undefined. The estimator in Section 4 jointly addresses the three challenges of quadratic parameter growth, endogeneity, and many candidate instruments.

values, so OLS can match the in-sample y values well without recovering the true coefficients.

4 Econometric Framework

4.1 Structural equation and identification

Fix an equation $i \in \{1, \dots, n_{\text{goods}}\}$ of the LA-AIDS system (10). Write $y \equiv w_i \in \mathbb{R}^T$ for the T -vector of budget shares for good i , and stack the constant, log real expenditure, and the n_{goods} log-prices into the regressor matrix $X \in \mathbb{R}^{T \times p_x}$ with $p_x = n_{\text{goods}} + 2$. Let $Z \in \mathbb{R}^{T \times p_z}$ be the matrix of instruments.¹⁵ Write $x_t \in \mathbb{R}^{p_x}$ and $z_t \in \mathbb{R}^{p_z}$ for the t -th rows of X and Z . The structural equation takes the form

$$y = X\beta + u, \quad (11)$$

where $\beta = (\alpha_i, \beta_i, \gamma_{i1}, \dots, \gamma_{i, n_{\text{goods}}})' \in \mathbb{R}^{p_x}$ collects the intercept, the expenditure coefficient, and the cross-price parameters for equation i , and $u \in \mathbb{R}^T$ is the structural error. The instrument matrix Z satisfies the conditional moment restriction

$$\mathbb{E}[u_t \mid z_t] = 0, \quad (12)$$

and is related to the regressors through the first-stage equation

$$X = Z\Pi + V, \quad \mathbb{E}[v_t \mid z_t] = 0, \quad (13)$$

where $\Pi \in \mathbb{R}^{p_z \times p_x}$ is the matrix of first-stage coefficients.

Identification of β requires the conditional moment restriction (12) and the order condition $p_z \geq p_x$. In the high-dimensional regime where both p_x and p_z can be large relative to T , I additionally assume sparsity: each column of Π has at most $s_A \ll p_z$ non-negligible entries and β has at most $s_\beta \ll p_x$ non-negligible entries.¹⁶

¹⁵In demand applications, valid instruments commonly include cost-shifters and lagged prices of related goods.

¹⁶Approximate sparsity requires that most entries are small rather than exactly zero. Gold et al. (2020) prove their results under ℓ_0 sparsity but note the extension to the approximate sparsity of Belloni et al. (2012).

4.2 Two-stage Lasso

The first stage estimates the conditional mean vector $d_t = \mathbb{E}[x_t \mid z_t]$ by separate Lasso regressions of each column of X on Z :

$$\hat{\alpha}_j \in \arg \min_{a \in \mathbb{R}^{p_z}} \left\{ \frac{\|x^j - Za\|_2^2}{2T} + r_j \|a\|_1 \right\}, \quad j = 1, \dots, p_x, \quad (14)$$

where $x^j \in \mathbb{R}^T$ is the j -th column of X and r_j is a tuning parameter. The predicted conditional means are $\hat{d}_{tj} = z_t' \hat{\alpha}_j$, assembled into $\hat{D} = Z\hat{\Pi} \in \mathbb{R}^{T \times p_x}$ where $\hat{\Pi}$ has columns $\hat{\alpha}_j$.

The second stage applies Lasso to the structural equation with $\hat{D}$ replacing the unobserved D :

$$\hat{\beta} \in \arg \min_{b \in \mathbb{R}^{p_x}} \left\{ \frac{\|y - \hat{D}b\|_2^2}{2T} + r_\beta \|b\|_1 \right\}. \quad (15)$$

Tuning parameters r_j and r_β are selected by cross-validation.

The initial estimator $\hat{\beta}$ is consistent under sparsity but does not have an asymptotic distribution suitable for inference (Knight and Fu (2000); Pötscher and Leeb (2009)). The ℓ_1 penalty produces shrinkage bias that does not vanish under the standard $\sqrt{T}$ scaling, so standard confidence intervals cannot be constructed from $\hat{\beta}$ directly.

4.3 Debiased one-step update

Following Gold et al. (2020), I reduce the bias in $\hat{\beta}$ via a one-step Newton-Raphson update targeting the empirical moment $\frac{\hat{D}'(y - X\hat{\beta})}{T} = 0$, the sample analog of the orthogonality condition $\mathbb{E}[d_t u_t] = 0$ that follows from (12). The update takes the form

$$\tilde{\beta}_j = \hat{\beta}_j + \frac{1}{T} \hat{\theta}_j' \hat{D}'(y - X\hat{\beta}), \quad j = 1, \dots, p_x, \quad (16)$$

where $\hat{\theta}_j \in \mathbb{R}^{p_x}$ is the j -th row of the estimated precision matrix $\hat{\Theta} \approx \Sigma_d^{-1}$ and $\Sigma_d = \mathbb{E}[d_t d_t']$ is the population covariance matrix of the conditional means.

Precision matrix estimation. In high dimensions the sample covariance $\hat{\Sigma}_d = \frac{\hat{D}'\hat{D}}{T}$ is ill-conditioned, and inverting it directly to obtain $\Theta = \Sigma_d^{-1}$ is unstable. I use the CLIME program of Cai et al. (2011), which targets a sparse Θ through a per-row linear program

rather than inverting $\hat{\Sigma}_d$:

$$\hat{\theta}_j \in \arg \min_{\theta \in \mathbb{R}^{p_x}} \|\theta\|_1 \quad \text{subject to} \quad \|\hat{\Sigma}_d \theta - e_j\|_\infty \leq \mu, \quad (17)$$

where $e_j \in \mathbb{R}^{p_x}$ is the j -th canonical basis vector and $\mu > 0$ is a tolerance parameter. Solving (17) row by row never inverts $\hat{\Sigma}_d$, and under sparsity CLIME is consistent for Θ even when p_x exceeds T . I set the tolerance at the theoretical convergence rate established by Cai et al. (2011):

$$\mu = \kappa \sqrt{\frac{\log p_x}{T}},$$

where $\kappa > 0$ controls the level. The order $\sqrt{\log p_x/T}$ is the rate at which μ must vanish for $\hat{\Theta}$ to attain the optimal convergence rate under sparsity. I use $\kappa = 1.2$, the value chosen in Gold et al. (2020)'s simulations, with Appendix B confirming that substantive conclusions are stable across $\kappa \in [0.5, 3.0]$.

4.4 Asymptotic normality

The update (16) mirrors a Newton-Raphson one-step in form, but classical one-step theory requires a $\sqrt{T}$ -consistent initial estimator and Lasso does not attain this rate. Gold et al. (2020) establish that the update is nevertheless $\sqrt{T}$ -consistent and asymptotically Gaussian under homoskedasticity. Caner and Kock (2018) establish an analogous result under conditional heteroskedasticity in a related framework, the case relevant for budget-share errors. Formally,

$$\frac{\sqrt{T}(\tilde{\beta}_j - \beta_j)}{\omega_j} \xrightarrow{d} \mathcal{N}(0, 1),$$

where ω_j^2 is the asymptotic variance.

4.5 Variance estimation and confidence intervals

The limit theorem above continues to hold when ω_j is replaced by any consistent estimator $\hat{\omega}_j$. I use the heteroskedasticity-robust sandwich,

$$\hat{\omega}_j^2 = \hat{\theta}'_j \left[\frac{1}{T} \sum_{t=1}^T \hat{u}_t^2 \hat{d}_t \hat{d}'_t \right] \hat{\theta}_j, \quad (18)$$

where $\hat{u}_t = y_t - x_t' \hat{\beta}$ are the structural residuals. Budget shares are bounded in $[0, 1]$, so the support of u_t is narrower when the share is near zero or one than when it is in the interior. The conditional variance of u_t is therefore highest at intermediate share levels. This makes the sandwich the correct estimator for AIDS estimation.¹⁷

An asymptotic $100(1 - \tau)\%$ confidence interval for β_j is

$$\tilde{\beta}_j \pm z_{\frac{\tau}{2}} \frac{\hat{\omega}_j}{\sqrt{T}}, \quad (19)$$

where $z_{\frac{\tau}{2}} = \Phi^{-1}(1 - \frac{\tau}{2})$ is the standard normal quantile.

4.6 Cross-fitting

The debiased update (16) uses $\hat{D}$ and $\hat{\beta}$, both estimated from the same sample as the structural residuals $\hat{u}$. In finite samples $\hat{D}$ is correlated with the structural shock u , since both are functions of the same observations, and this dependence propagates to $\hat{\beta}$ and to the residuals. The debiased update inherits this dependence. This can add a finite-sample bias separate from the shrinkage bias that (16) removes asymptotically, and can contaminate the sandwich variance (18), distorting finite-sample coverage in principle. I address this by K -fold cross-fitting following Chernozhukov et al. (2018).

Partition $\{1, \dots, T\}$ into K folds $\mathcal{F}_1, \dots, \mathcal{F}_K$ of approximately equal size. For each fold k , fit the first-stage Lasso on the data outside fold k :

$$\hat{\alpha}_j^{(-k)} \in \arg \min_{a \in \mathbb{R}^{p_z}} \left\{ \frac{1}{2(T - |\mathcal{F}_k|)} \sum_{t \notin \mathcal{F}_k} (x_{tj} - z_t' a)^2 + r_j \|a\|_1 \right\}, \quad (20)$$

and use the fitted model to predict on the held-out fold, setting $\hat{d}_{tj}^{CF} = z_t' \hat{\alpha}_j^{(-k)}$ for $t \in \mathcal{F}_k$. Stacking the out-of-fold predictions gives the cross-fitted matrix $\hat{D}^{CF}$, which replaces $\hat{D}$ everywhere in (15)–(16). Each prediction uses only data from outside its fold, so $\hat{D}_t^{CF}$ does not depend on u_t , which breaks the within-sample dependence by construction. I implement $K = 5$ as the default and report non-cross-fitted results alongside to assess whether this dependence is the binding finite-sample issue for coverage.

¹⁷Under homoskedasticity, the sandwich is asymptotically equivalent to the homoskedastic variance estimator of Gold et al. (2020), $\hat{\sigma}_u^2 \hat{\Theta}_{jj}$ with $\hat{\sigma}_u^2 = (\frac{1}{T}) \sum_t \hat{u}_t^2$.

5 Numerical experiments

In this section I present a Monte Carlo simulation study of the finite-sample properties of the estimator developed in Section 4, which I refer to as *debiased HD-IV*. The design scales the number of goods n_{goods} and the number of candidate instruments p_z relative to a fixed sample size T , evaluating debiased HD-IV against five comparators on RMSE and empirical coverage of nominal 95% confidence intervals. The estimands are the Engel-curve slope β and the own-price coefficient γ_{ii} for a single AIDS share equation.¹⁸

I estimate each equation separately, without enforcing cross-equation restrictions during estimation and without imposing Slutsky symmetry on the recovered coefficients.¹⁹ Results are averaged over $R = 100$ replications per grid point.²⁰

5.1 Data-generating process

Log prices for good j are generated as

$$\log p_{jt} = k(z_t' \pi_j) + \rho_u u_t + v_{jt}, \quad (21)$$

where $z_t \in \mathbb{R}^{p_z}$ is i.i.d. standard normal, $\pi_j \in \mathbb{R}^{p_z}$ has $s_A = 5$ nonzero entries chosen at random, $k > 0$ governs instrument strength, ρ_u controls the endogeneity of prices through the structural error u_t , and v_{jt} is idiosyncratic noise. A common price factor with correlation $\rho_{\text{col}} = 0.95$ reproduces the high collinearity typical of demand data. The structural error u_t and the first-stage residuals v_{jt} are jointly Gaussian, with prices endogenous when $\rho_u \neq 0$. Total expenditure $\log\left(\frac{x_t}{P_t^*}\right)$ is treated as exogenous in the main analysis. Instrumenting expenditure jointly with prices is left for future work.

The structural share equation is given in (10). The true parameter vector $(\alpha_i, \beta_i, \gamma_i)$ is drawn once at the start of each experiment and held fixed across replications. Increases in n_{goods} and p_z therefore scale estimation difficulty rather than the target. The true

¹⁸Code to reproduce the simulations and all figures is available in Appendix C.

¹⁹The adding-up, homogeneity, and Slutsky symmetry restrictions hold at the data-generating process by construction (Section 5.1). Joint estimation of the share system with these restrictions imposed would sharpen the estimates and is the natural next step, but is outside the scope of the present methodological contribution.

²⁰At this replication count, the Monte Carlo standard error on a coverage estimate is about two percentage points, so small differences in coverage between estimators may reflect sampling noise rather than a real gap.

parameters satisfy the adding-up (6), homogeneity (7), and Slutsky symmetry (8) restrictions by construction. The cross-price vector γ_i is approximately sparse: a small number of structurally large entries capture substitutes, and the remaining entries are small in magnitude rather than exactly zero. The homogeneity restriction $\sum_j \gamma_{ij} = 0$ forces each row of price coefficients to sum to zero, so the would-be-zero entries are shifted by a small constant to balance the structurally large ones. The debiased framework accommodates such approximately sparse vectors, as noted in Gold et al. (2020) and developed in Belloni et al. (2012).

By contrast, the first stage in this DGP is exactly sparse: each π_j has $s_A = 5$ nonzero entries by construction, and there is no analog of the homogeneity restriction to perturb the zeros. A first-stage selection diagnostic at $n_{\text{goods}} = 100$ illustrates how the Lasso behaves under this structure. Cross-validated Lasso recovers most of the structurally large entries of the true coefficient vector but also keeps a number of spurious entries.²¹ Debiased HD-IV does not require the Lasso to have selected the true support. The one-step correction adjusts the targeted coefficient through the precision matrix regardless of which entries Lasso keeps.

5.2 Scenarios and grid

I consider three scenarios. Scenario A sets $\rho_u = 0$, so prices are exogenous and OLS is consistent. Scenario B sets $\rho_u = 0.6$ with $k = 1.0$, giving endogeneity and strong instruments. Scenario C sets $\rho_u = 0.6$ with $k = 0.3$, giving endogeneity and weak instruments.

The main sweep fixes $T = 300$ and varies $n_{\text{goods}} \in \{50, 100, 150, 200, 250, 300\}$, so the structural dimension $p_x = n_{\text{goods}} + 2$ ranges from well below T to just above it at the upper grid point. The candidate instrument count is held at $p_z = 350$, sufficient to satisfy the order condition $p_z \geq p_x$ at every grid point. A separate p_z -sweep holds $n_{\text{goods}} = 100$ fixed and varies $p_z \in \{110, 200, 300, 400, 500\}$ to isolate the many-instruments dimension.

All Lasso tuning is by 5-fold cross-validation. Standard errors use the heteroskedasticity-

²¹Bühlmann and Van De Geer (2011) discuss over-selection as typical behavior of cross-validated Lasso. CV selects the penalty for prediction accuracy, which keeps small but non-zero coefficients rather than zeroing them out.

robust sandwich form of (18). CLIME tuning fixes $\kappa = 1.2$ in $\mu = \kappa\sqrt{\log p_x/T}$, following Gold et al. (2020).

5.3 Estimators

Six estimators are compared in the main paper, with three further variants reported as robustness in Appendix B.

OLS is the no-correction baseline. It ignores endogeneity entirely and shows how a naive estimator behaves as dimension grows. Classical 2SLS uses all p_z candidate instruments. When $p_z > T$ the first-stage projection is rank-deficient and 2SLS reduces to OLS in finite samples. 2SLS with Lasso first stage (2SLS-LassoFS) of Belloni et al. (2012) Lasso-regresses each endogenous price on the candidate instruments and uses the fitted values in the second stage. The Lasso first stage remains well-defined when $p_z > T$. Post-Lasso IV of Belloni et al. (2012) instead uses Lasso for instrument selection only. It refits the first stage by OLS on the selected support, then passes the unshrunk fitted values to a classical 2SLS second stage. This is the comparator that Gold et al. (2020) benchmark against, and I include it for direct comparison with their results. Debiased HD-IV is the estimator introduced in Section 4. Debiased HD-IV with cross-fitting applies $K = 5$ -fold cross-fitting to the first-stage Lasso following Chernozhukov et al. (2018), as a finite-sample refinement (see Section 4.6). The cross-fitting partition is separate from the cross-validation used to tune the Lasso penalty.²²

At $n_{\text{goods}} = 300$ there are more regressors than time periods ($p_x = 302 > T = 300$), so the Gram matrix $\hat{\Sigma}_d = \frac{1}{T}\hat{D}'\hat{D}$ is singular. None of the three precision-matrix constructions inverts $\hat{\Sigma}_d$ itself, and none carries a finite-sample guarantee; at this point they differ in computational cost. Mainline CLIME solves a separate linear program (17) for each of the p_x coordinates, and some of those programs are tight enough that the tolerance must be widened before they are feasible, which is the expensive step of the estimator here.²³ The ridge-regularized variant needs only one inversion of the full-rank, well-conditioned lift $\hat{\Sigma}_d + \lambda I$, at the cost of a small regularization bias, and the nodewise-Lasso variant needs

²²Three further high-dimensional variants are reported as estimator-level robustness in Appendix B: the elastic-net first stage of Bae (2018), and the ridge and nodewise-Lasso precision-matrix variants.

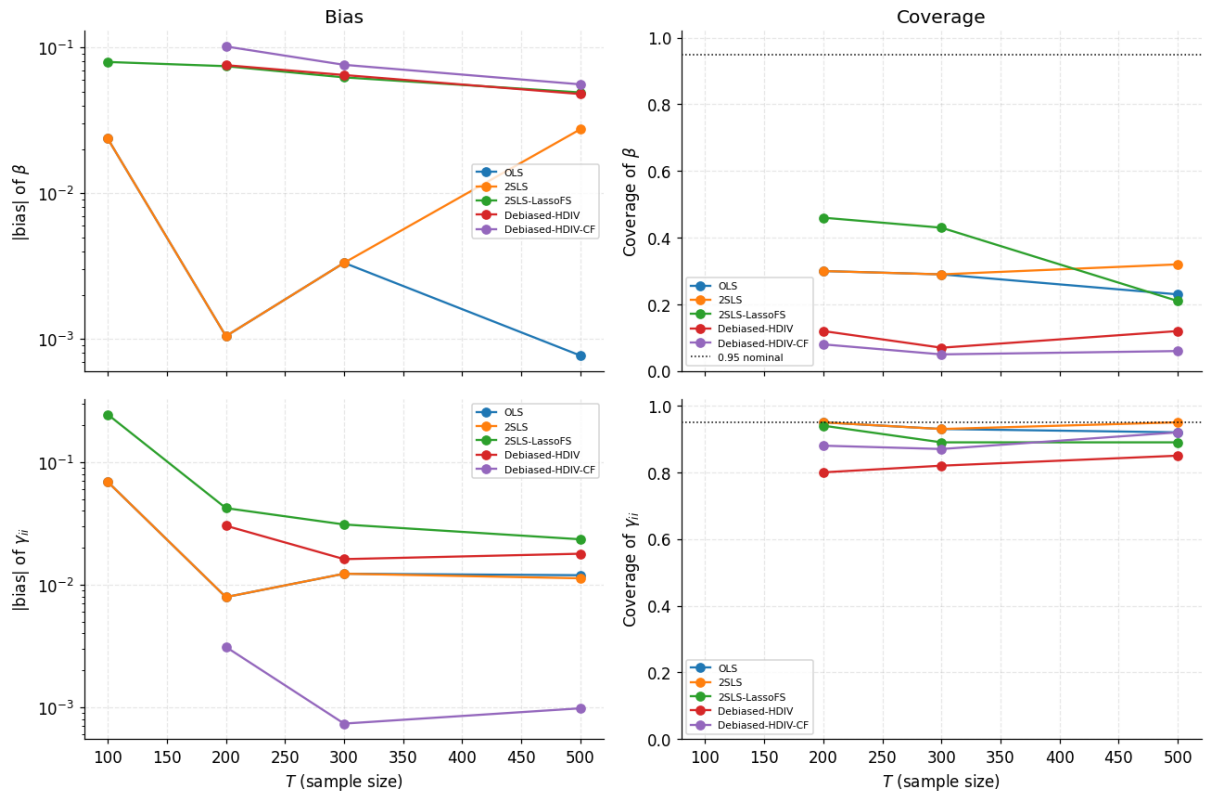
²³Solving (17) separately for every row, across all Monte Carlo replications, is what makes the mainline estimator costly in this regime.

only a set of column-wise sparse regressions, which stay well-defined for any p_x relative to T . I therefore report mainline CLIME as missing at this grid point and use the two cheaper constructions to extend the estimator through the boundary. This is specific to the mainline CLIME construction, not a problem with the debiased framework.

5.4 Asymptotic validation

I first verify that the asymptotic guarantees of Gold et al. (2020) hold in the AIDS DGP. Theory predicts that as T grows, bias for the debiased estimators should decay toward zero and empirical coverage should approach the nominal level. Figure 2 tests this by plotting absolute bias and empirical coverage of β and γ_{ii} against $T \in \{100, 200, 300, 500\}$ at fixed $n_{\text{goods}} = 100$ in Scenario B.

Figure 2. Asymptotic validation: bias and coverage versus T



Notes: $T \in \{100, 200, 300, 500\}$ at fixed $n_{\text{goods}} = 100$, $p_z = 350$, Scenario B, $R = 100$. Top row β , bottom row γ_{ii} . Left column absolute bias on log scale, right column empirical coverage of 95% confidence intervals; the dashed line marks the nominal level at 0.95. At $T = 100$ the structural design matrix is rank-deficient ($p_x = 102 > T$), so the asymptotic-convergence comparison should be read from $T = 200$ onward. Some estimators are not plotted at $T = 100$.

The picture for γ_{ii} matches the theoretical prediction. Bias for cross-fitted debiased HD-IV decays fastest, reaching near-zero values from $T = 300$ onward. Mainline debiased HD-IV decays more slowly. Coverage for both debiased variants starts below the nominal level and improves as T grows. The cross-fitted variant approaches nominal by $T = 500$. The mainline variant improves more slowly and remains somewhat below at $T = 500$. The cross-fitted variant outperforms the mainline for the reasons discussed in Section 4.6.

The picture for β only partially matches the prediction. Bias for the debiased variants decreases modestly as T grows, but coverage does not approach the nominal level over the range tested. Cross-fitting does not improve β coverage, which rules out the in-sample dependence between the first-stage Lasso and the structural residuals as the binding issue. The pattern instead reflects a documented finite-sample limitation of debiased Lasso for coefficients whose magnitude is small relative to the Lasso shrinkage scale.²⁴ The true β values in this DGP are drawn small, putting them in this regime.²⁵ OLS β coverage is also below nominal, but the bias panel shows OLS β bias is small. The OLS coverage failure is a standard-error problem rather than a bias problem.²⁶

5.5 Results

Scenario A is a calibration check. With $\rho_u = 0$ the endogeneity correction is unnecessary and all estimators should recover the structural coefficients comparably. Full results across all nine estimators are in Appendix B.2. The qualitative picture matches expectations. At low n_{goods} all estimators have comparable RMSE. As n_{goods} approaches T the unregularized estimators deteriorate because the design matrix becomes ill-conditioned. At the boundary in Scenario A, only the ridge-regularized debiased variant maintains the

²⁴Van de Geer et al. (2014) note that the one-step correction is asymptotically valid but in finite samples leaves a residual bias when the true coefficient is small. Lasso shrinks small coefficients hard, and the correction does not fully undo the shrinkage at moderate T . The bias should vanish as T grows, but slowly. The asymptotic regime is not reached at the T values tested here.

²⁵Calibrating β small is empirically realistic. Budget-share responses to log real expenditure are typically small in demand data (Deaton & Muellbauer, 1980), and the AIDS adding-up restriction $\sum_i \beta_i = 0$ forces the β values to balance around zero. The small- β setting here matches the regime AIDS faces in practice.

²⁶The Stone deflator $\log P_t^* = \sum_j w_{jt} \log p_{jt}$ uses observed shares as weights, which are jointly determined with the structural error through the share equation. This creates a simultaneity for OLS in principle, though the bias is small in this simulation (the bias panel shows OLS β bias near zero). The dependence still affects the sampling variability of the OLS estimator beyond what the heteroskedasticity-robust standard error captures.

RMSE level it held below the boundary. Ridge handles rank deficiency directly through the λI term and serves as the boundary fallback. The other debiased variants are more sensitive to rank deficiency and deteriorate at the boundary.²⁷ The debiased variants also undercover for β in Scenario A, though less severely than in Scenario B. This confirms that the β coverage gap reflects the small-coefficient limitation of debiased Lasso rather than the endogeneity setting.

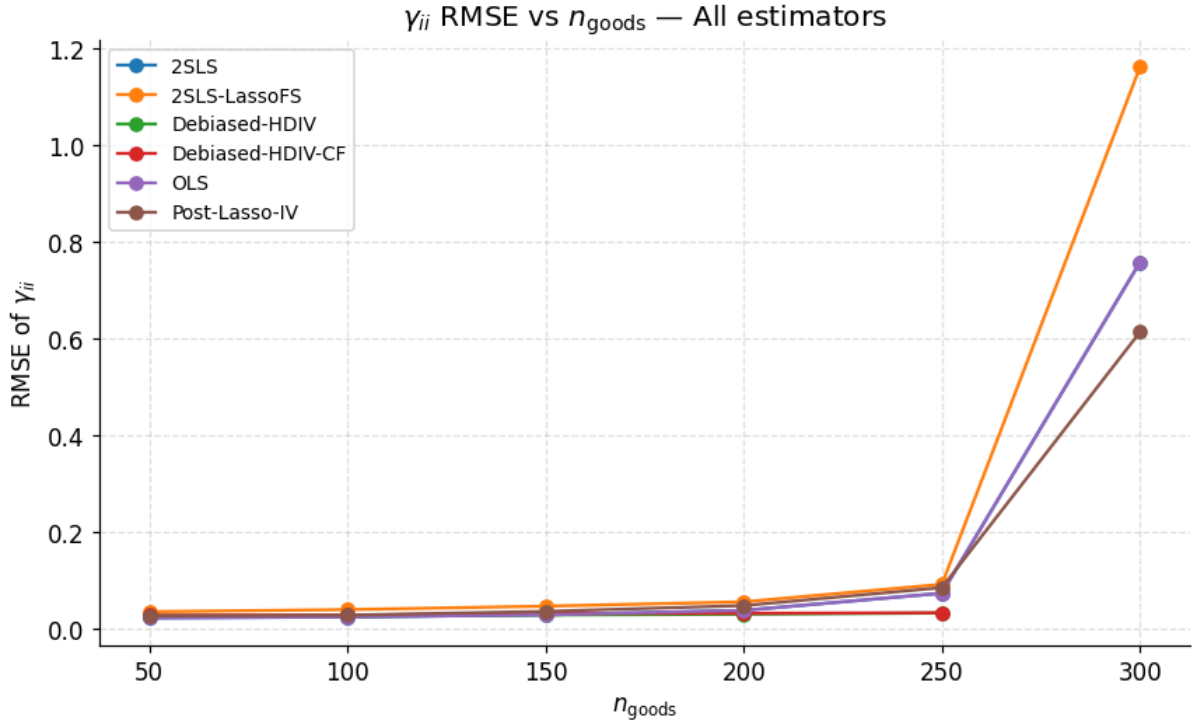
5.5.1 Scenario B: main results

Scenario B is the case of interest. Two structural features hold across the Scenario B figures that follow. First, mainline debiased HD-IV and its cross-fitted variant overlap closely on γ_{ii} throughout the inner grid at fixed $T = 300$. The cross-fitting advantage scales with T and shows clearly in the T -sweep of Figure 2, but at fixed $T = 300$ across n_{goods} the two are visually indistinguishable on γ_{ii} . On β coverage the variants separate visibly, with cross-fitting slightly worse than mainline, as discussed below. Second, both debiased CLIME variants are reported missing at the boundary $n_{\text{goods}} = 300$, for the reasons given above. The framework’s extension through the boundary is delivered by the ridge and nodewise precision-matrix variants reported in Appendix B. The precision-matrix robustness check in Figure 26 shows those variants extending to $n_{\text{goods}} = 300$ with RMSE values comparable to the mainline on the inner grid.

Figure 3 reports RMSE of estimators of γ_{ii} across the n_{goods} -grid.

²⁷This does not make ridge the better choice overall. CLIME exploits the assumed sparsity of the inverse to give sharper estimates in the typical full-rank regime, where it is the mainline choice.

Figure 3. RMSE of γ_{ii} , Scenario B

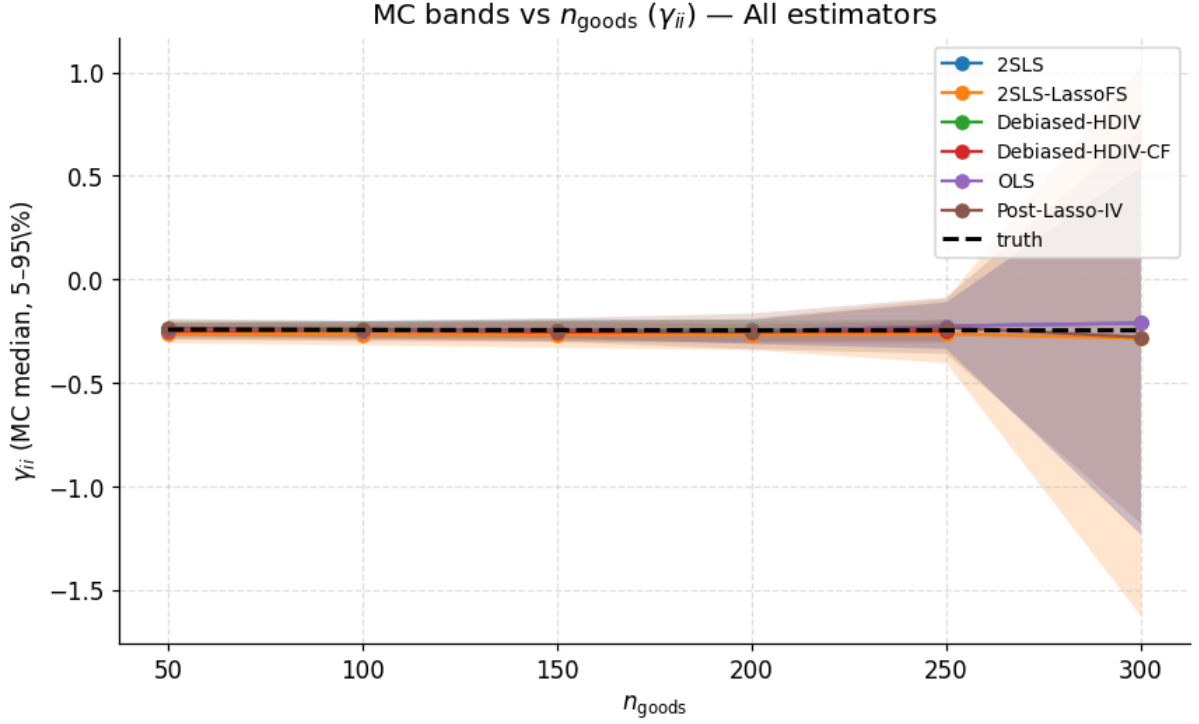


Notes: $n_{\text{goods}} \in \{50, 100, 150, 200, 250, 300\}$, $T = 300$, $p_z = 350$, Scenario B ($\rho_u = 0.6$, $k = 1.0$), $R = 100$.

Across most of the inner grid the six estimators have comparable RMSE on γ_{ii} and the lines overlap. At $n_{\text{goods}} = 250$ the comparators begin to separate from the debiased lines: 2SLS-LassoFS and OLS tick up modestly while the debiased variants stay flat. At the boundary $n_{\text{goods}} = 300$ the comparators all spike: 2SLS-LassoFS most sharply, then OLS (which classical 2SLS reduces to in this regime) and Post-Lasso IV.

Figure 4 shows Monte Carlo bands for γ_{ii} . The bands separate bias from variance, which the headline RMSE figure combines into a single number.

Figure 4. Monte Carlo bands for γ_{ii} , Scenario B

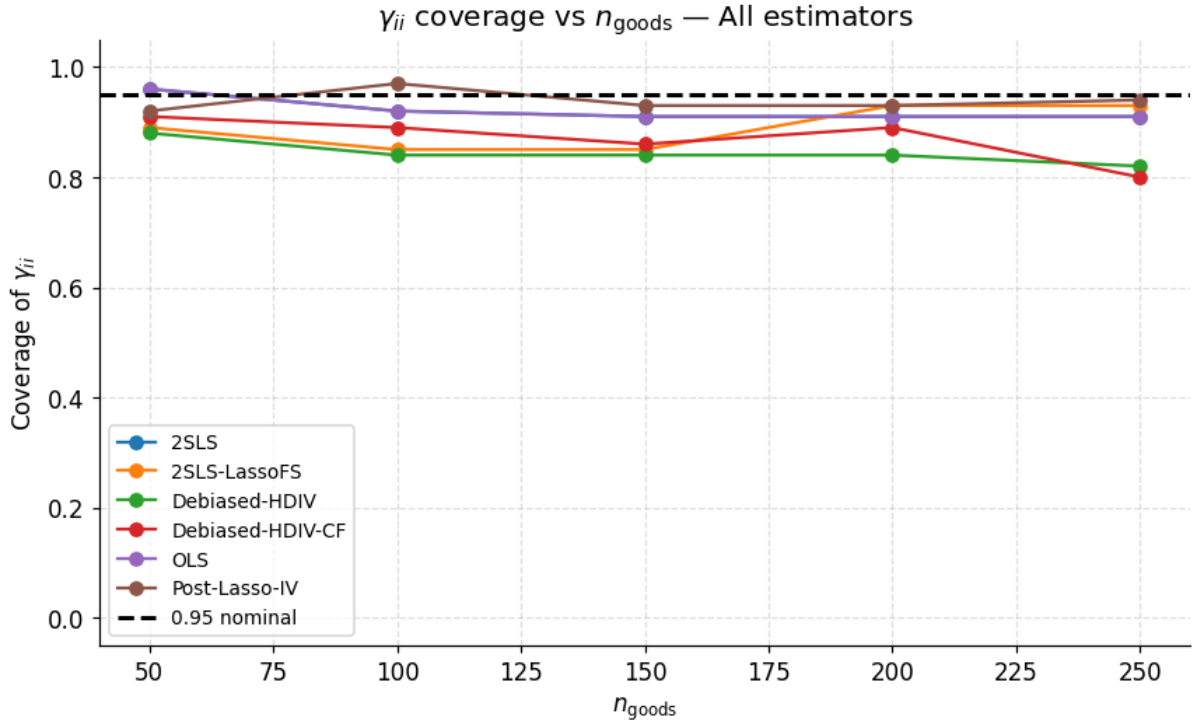


Notes: Median (lines) and 5–95% bands (shaded) for γ_{ii} across the n_{goods} -grid at $T = 300$, $p_z = 350$, Scenario B, $R = 100$. The dashed black line marks the true parameter.

All estimator medians sit at the truth across the inner grid with tight 5–95% bands. At the boundary, the comparator bands widen, most dramatically for 2SLS-LassoFS, with OLS and Post-Lasso IV showing modest widening. The comparator medians remain close to the truth, so their boundary RMSE spike in Figure 3 reflects variance rather than bias. The rank-deficient design matrix at the boundary gives the unregularized estimators explosive sampling variance, which RMSE picks up even when medians stay near the truth.

Figure 5 reports empirical coverage of γ_{ii} across the n_{goods} -grid.

Figure 5. Coverage of γ_{ii} , Scenario B

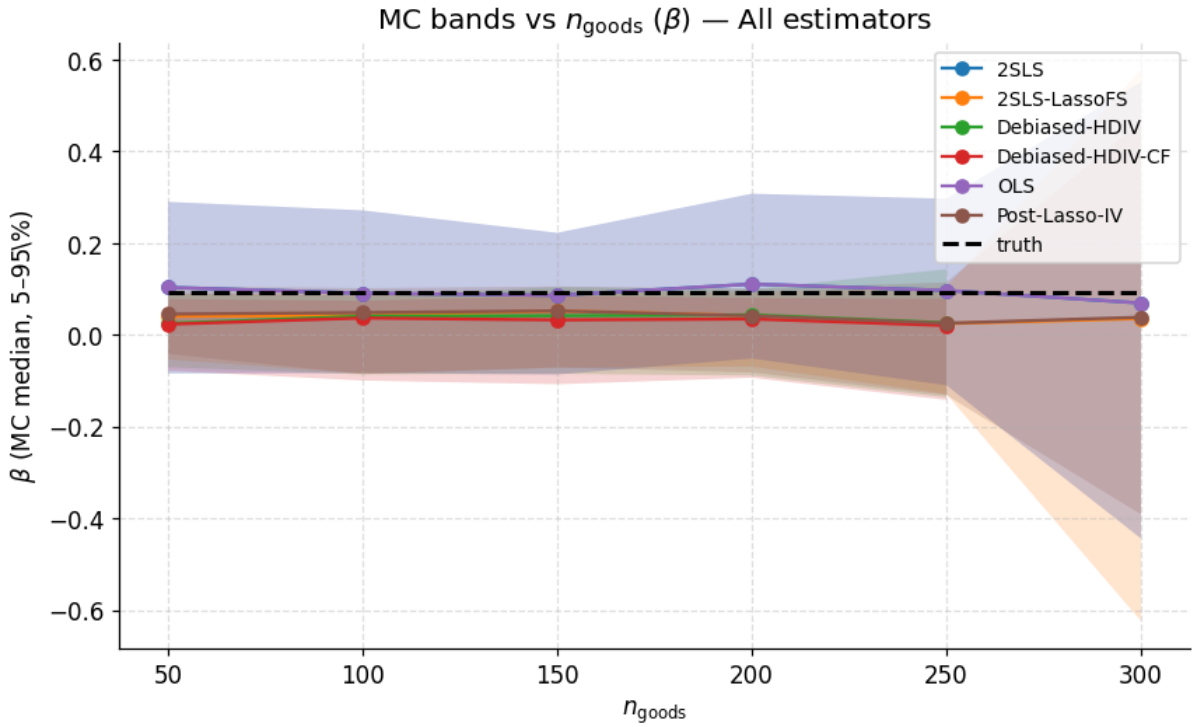


Notes: Empirical coverage of nominal 95% confidence intervals for γ_{ii} across the n_{goods} -grid at $T = 300$, $p_z = 350$, Scenario B, $R = 100$. Dashed line marks the nominal level.

Coverage of γ_{ii} sits at or modestly below the nominal level across the grid for all estimators, with the debiased variants at the low end of the range.

For β , Figure 6 shows Monte Carlo bands, which visualize the finite-sample bias the debiased estimators carry on this coefficient.

Figure 6. Monte Carlo bands for β , Scenario B

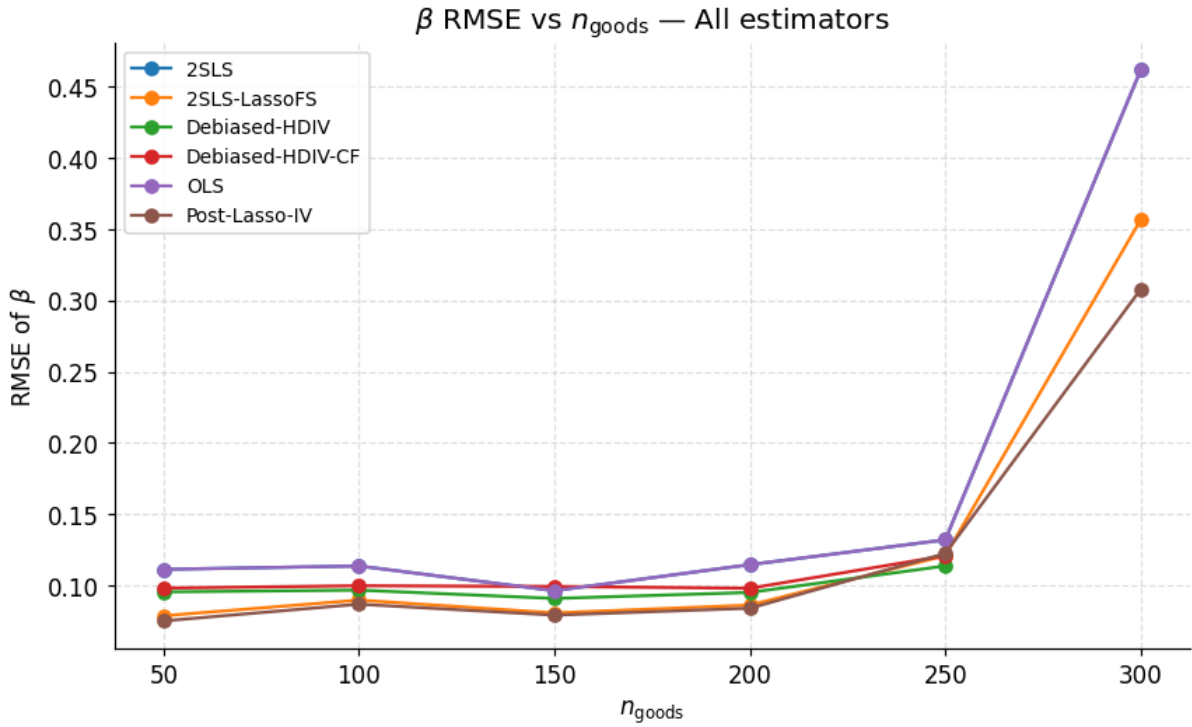


Notes: Median (lines) and 5–95% bands (shaded) for β across the n_{goods} -grid at $T = 300$, $p_z = 350$, Scenario B, $R = 100$. The dashed black line marks the true parameter.

The OLS median sits on the truth across the grid (bias on β is small for OLS because endogeneity passes to β only indirectly through the Stone-index regressor). The debiased medians sit below the truth by a small amount that is roughly constant across the grid. This small persistent bias, combined with debiased CLIME standard errors that turn out to be tight relative to the bias, drives the coverage gap for β documented below.

Figure 7 reports RMSE of β .

Figure 7. RMSE of β , Scenario B

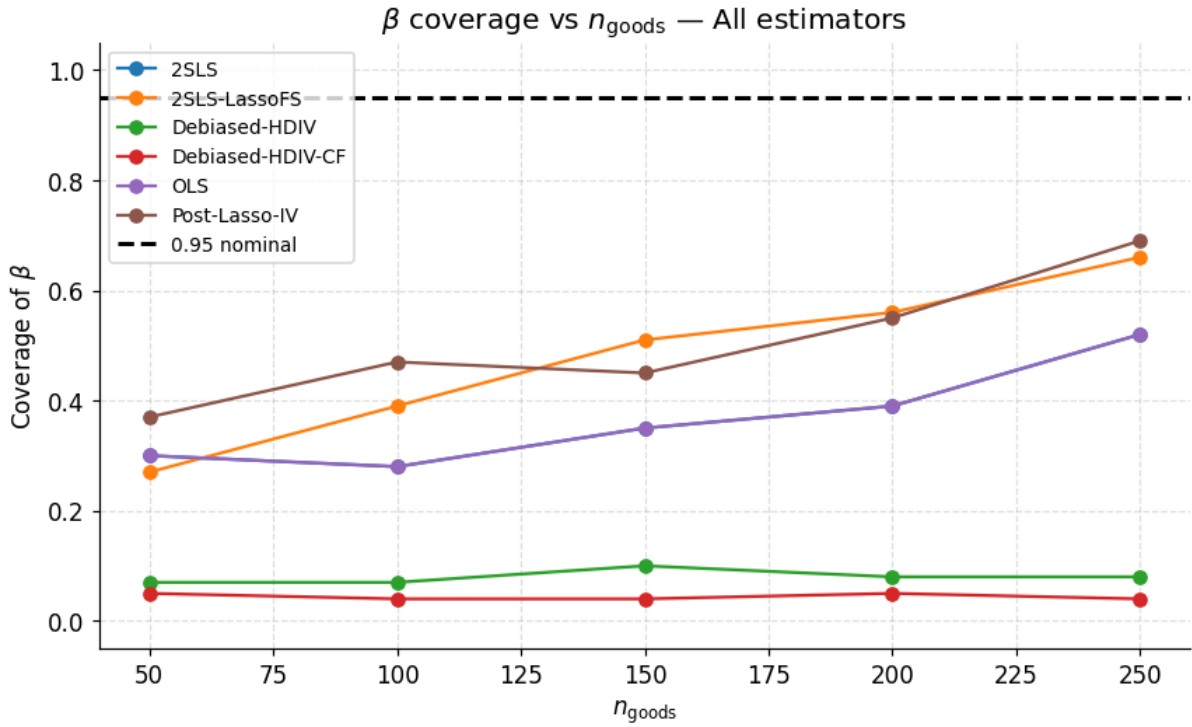


Notes: RMSE of β across the n_{goods} -grid at $T = 300$, $p_z = 350$, Scenario B, $R = 100$.

RMSE on β is similar across estimators through the inner grid, with the debiased estimators slightly lower than OLS. The debiased variants carry a small bias but tighter variance, while OLS is essentially unbiased but more variable. The lower variance of the debiased estimators outweighs their small bias in the RMSE comparison. At the boundary, OLS, 2SLS-LassoFS, and Post-Lasso IV jump sharply, mirroring the boundary behavior of γ_{ii} .

Figure 8 reports empirical coverage of β .

Figure 8. Coverage of β , Scenario B

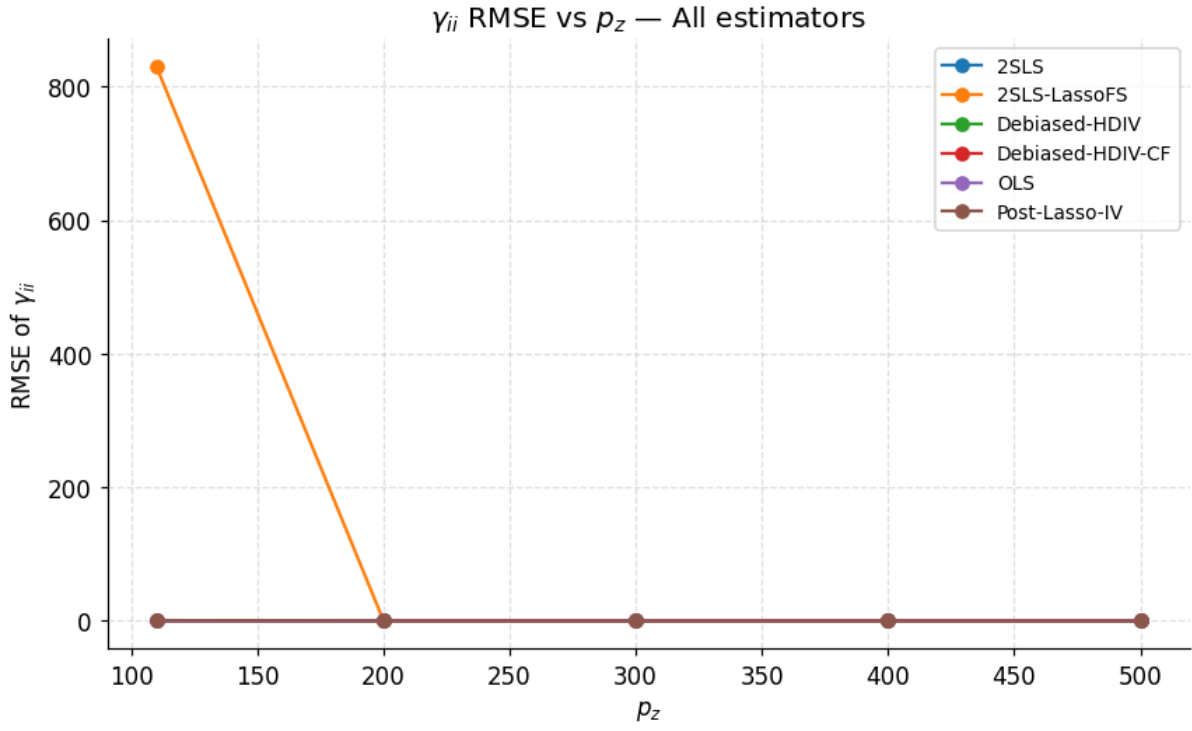


Notes: Empirical coverage of nominal 95% confidence intervals for β across the n_{goods} -grid at $T = 300$, $p_z = 350$, Scenario B, $R = 100$. Dashed line marks the nominal level.

All estimators undercover. The debiased variants undercover most severely, with empirical coverage well below nominal across the grid. Cross-fitting does not close the gap, as documented in Section 5.4. The pattern reflects the small-coefficient finite-sample limitation of debiased Lasso. A small persistent downward bias on β (visible in Figure 6) combined with debiased CLIME standard errors that are tight relative to this bias leaves the confidence intervals too narrow to catch the truth across the grid.

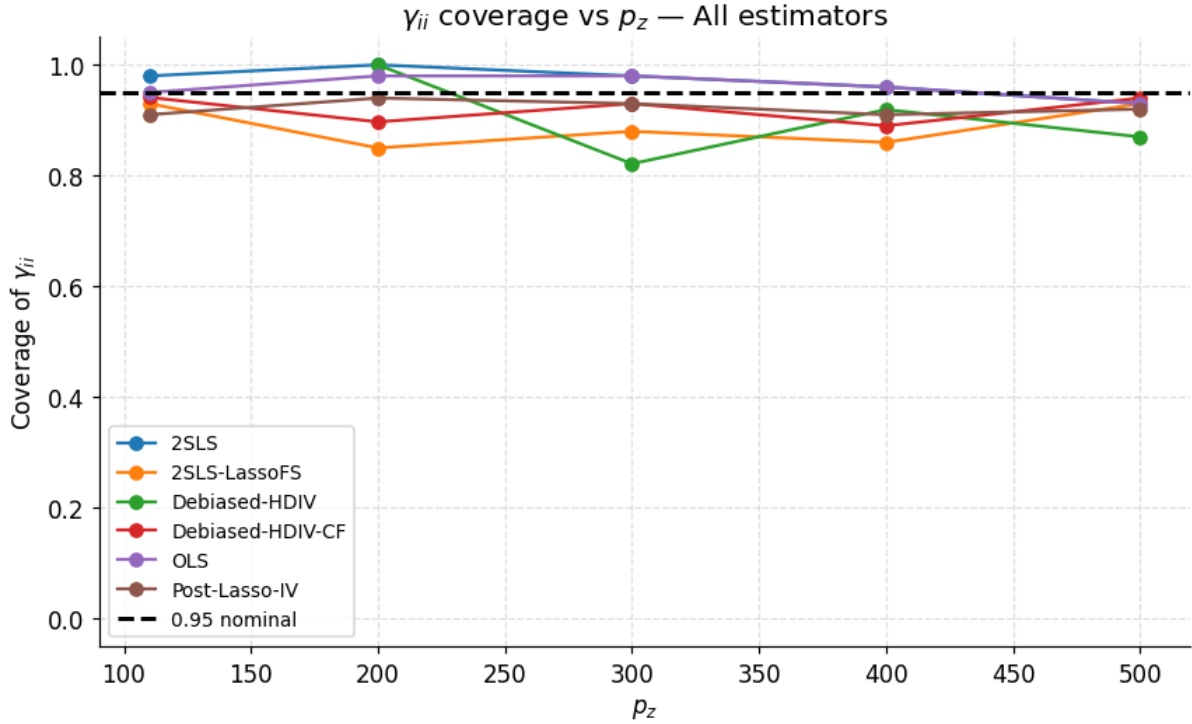
Figures 9 and 10 report the p_z -sweep at fixed $n_{\text{goods}} = 100$ for γ_{ii} . The small-coefficient limitation documented above for β does not depend on the instrument count, so β is not separately reported in the p_z -sweep.

Figure 9. RMSE of γ_{ii} in the p_z -sweep, Scenario B



Notes: RMSE of γ_{ii} across $p_z \in \{110, 200, 300, 400, 500\}$ at fixed $n_{\text{goods}} = 100$, $T = 300$, Scenario B, $R = 100$. Classical 2SLS reduces numerically to OLS once $p_z > T$.

Figure 10. Coverage of γ_{ii} in the p_z -sweep, Scenario B



Notes: Empirical coverage of nominal 95% confidence intervals for γ_{ii} across the p_z -sweep at fixed $n_{\text{goods}} = 100$, $T = 300$, Scenario B, $R = 100$. Dashed line marks the nominal level.

In the RMSE panel, 2SLS-LassoFS spikes at $p_z = 110$, where the candidate instrument count just satisfies the order condition but the Lasso first stage has too little structure to discipline the predictions. The spike disappears at $p_z = 200$ and beyond. The remaining five estimators are stacked tightly together at the bottom of the figure, indistinguishable on the scale set by the 2SLS-LassoFS spike. What can be read off the figure is that none of these five exhibits a comparable RMSE spike at low p_z or across the rest of the grid. Coverage of γ_{ii} in the companion figure stays close to the nominal level across p_z for all estimators (the coverage axis is bounded, so the lines are distinguishable there), with the debiased variants slightly below. The instrument dimension does not break any of the estimators in this sweep. Structural dimension is held at $p_x = 102 \ll T = 300$, so classical 2SLS reduces to OLS once $p_z > T$ without further damage, and the Lasso first stage selects the few informative instruments regardless of how many candidates are added.

5.5.2 Scenario C: weak instruments

Under weak identification (Scenario C), the Lasso first-stage variants break down. Detailed results are in Appendix B.3. In the n_{goods} -sweep, 2SLS-LassoFS and 2SLS-EnetFS RMSE spike by several orders of magnitude at the boundary. In the p_z -sweep, the same two estimators spike at $p_z = 110$, where the candidate instrument count just satisfies the order condition. OLS is unaffected by the weakness of the instruments since it does not use them, and its boundary RMSE spike is dominated on the figure scale by the Lasso-FS spikes. Post-Lasso IV is stable in RMSE but severely undercovers across the p_z -grid; under weak identification, the OLS-refit standard errors do not reflect the noise in the Lasso-selected first stage, leaving the confidence intervals too narrow. The debiased variants are more stable than the alternatives on RMSE but have their own coverage issues. The nodewise-Lasso variant undercovers severely at higher n_{goods} ; the column-by-column Lasso regressions that build its precision matrix are noisy under weak identification, and the noise grows with the matrix dimension. Weak identification is not a problem the debiased framework solves. Its value under weak instruments is more stable point estimation than the Lasso-FS comparators provide.

5.6 Robustness

Appendix B reports robustness checks along several dimensions. These include sensitivity of the debiased estimator to the CLIME tuning parameter κ , the ridge-regularized and nodewise-Lasso precision-matrix variants that extend the estimator through the boundary $p_x \geq T$ where mainline CLIME is not computed, and the elastic-net first-stage variant of Bae (2018). The substantive findings of this section are not driven by any of these choices. The appendix also reports the full sets of Scenario A and Scenario C results across all nine estimators.

6 Conclusion

This paper applies the debiased high-dimensional IV framework of Gold et al. (2020) to the Almost Ideal Demand System with many endogenous prices and many candidate

instruments. The Monte Carlo design compares debiased HD-IV against OLS, classical 2SLS, two Lasso-based IV variants of Belloni et al. (2012) (2SLS with a Lasso first stage and Post-Lasso IV), across three scenarios spanning exogenous prices, endogenous prices with strong instruments, and endogenous prices with weak instruments. The simulation establishes three findings.

First, for the own-price coefficient γ_{ii} , the debiased framework remains stable as n_{goods} approaches T while the comparators degrade. In the scenario with endogenous prices and strong instruments, RMSE on γ_{ii} is comparable across all six estimators when n_{goods} is well below T . As n_{goods} approaches T the comparators separate from the debiased lines and spike sharply at the boundary, while the debiased variants hold their level up to it. Coverage of γ_{ii} sits modestly below the nominal level for the debiased variants across the grid.

Second, the methodology extends through the rank-deficient regime $p_x \geq T$ via ridge-regularized and nodewise-Lasso variants of the precision matrix. These variants recover the same qualitative picture as the mainline estimator on the inner grid, and they extend the practical range of the framework into the rank-deficient regime $p_x \geq T$, where mainline CLIME is costly and therefore not computed.

Third, debiased coverage for the Engel-curve slope β is poor in finite samples and remains poor at the largest T examined. The true β values are small relative to the Lasso shrinkage scale, leaving a residual bias that combines with tight CLIME-based standard errors to produce coverage misses across all debiased variants. Cross-fitting does not close this gap. This is the documented small-coefficient limitation of debiased Lasso, present in non-IV settings and not specific to AIDS. It is not a dimensionality failure. The cross-price coefficients γ_{ij} that make AIDS high-dimensional are recovered acceptably by the same estimator as n_{goods} scales, and the bias appears only on the small-magnitude Engel slope β .

For applied work the practical guidance is targeted. Debiased HD-IV is the right tool for inference on individual cross-price elasticities at high dimension, where it delivers comparable or better RMSE than the alternatives and coverage on γ_{ii} that sits modestly below the nominal level. Inference on the Engel-curve slope warrants more caution at the

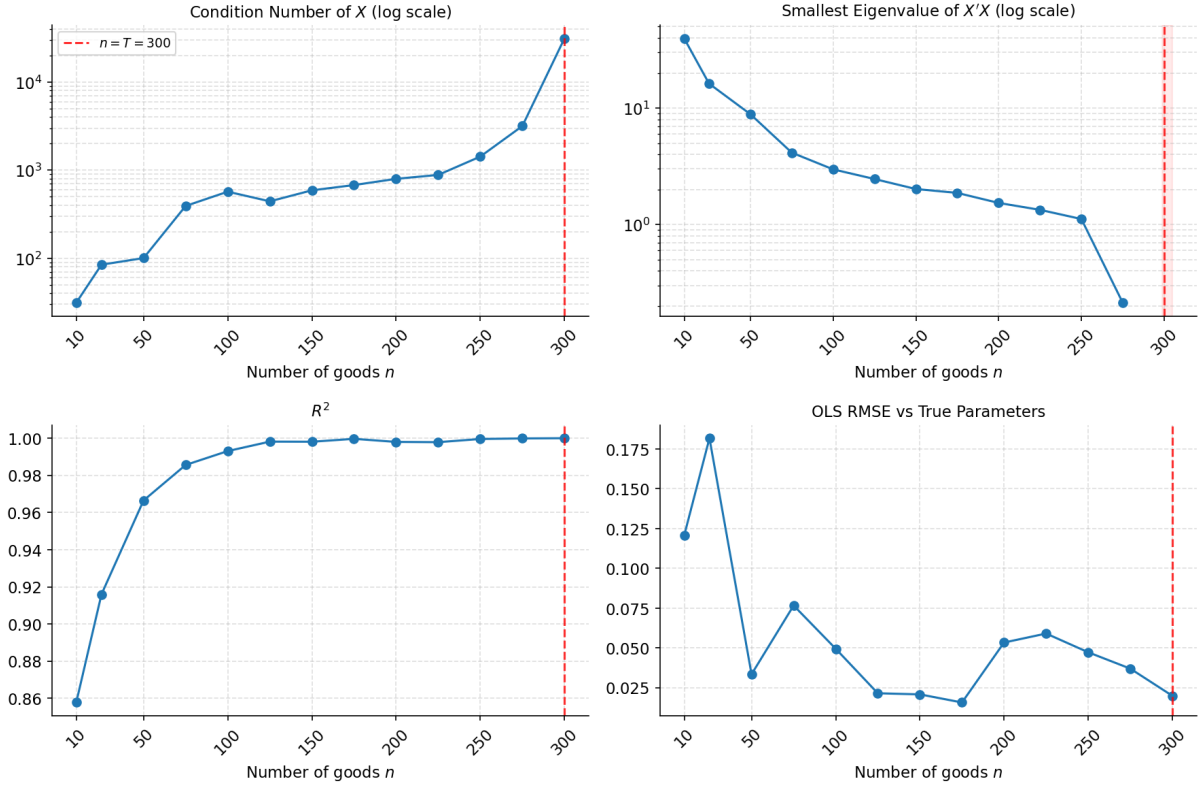
sample sizes considered here. Applied researchers may wish to pair the debiased point estimate with standard errors from a less-shrinking estimator or use bootstrap intervals for β .

Several natural extensions are left for future work. The most direct is joint estimation of the AIDS share system with cross-equation restrictions imposed during estimation, which may also tighten β coverage by tying values across equations. Another is an empirical application to scanner data or product-level trade data. A third is treatment of total expenditure as endogenous and instrumented alongside prices. A fourth, more methodological, is to combine the debiased construction here with the automatic-debiasing machinery of Chernozhukov et al. (2022) or the deep-learning approach of Farrell et al. (2021), allowing a more flexible first stage than the linear-sparse specification used here.

A Additional dimensionality diagnostics

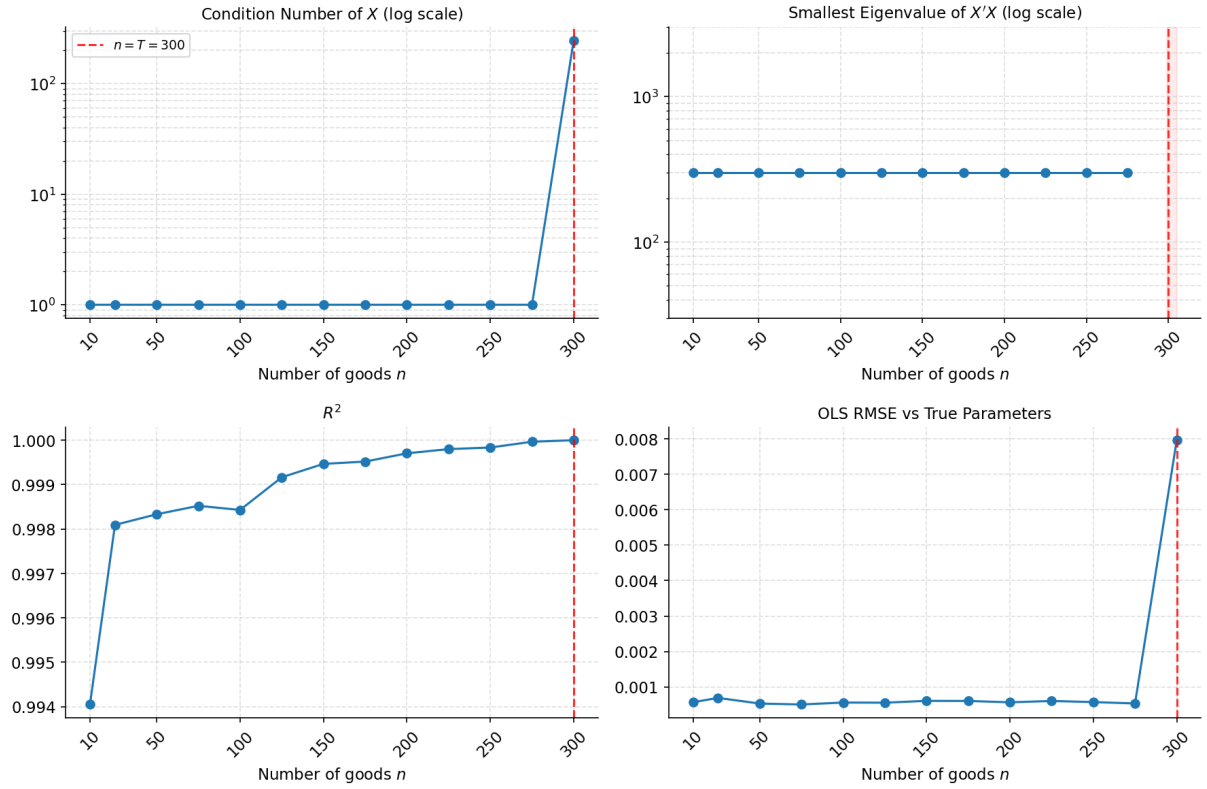
This appendix contains the supporting figures referenced in Section 3.3. The simulation design and DGP are identical to those of Figure 1. The first two figures vary the price-generation mechanism, using independent AR(1) prices and orthogonal regressors, to isolate the contribution of price collinearity. The remaining figures report additional diagnostics under correlated prices. These include perturbation sensitivity, which measures stability of the OLS solution under small data perturbations, and Monte Carlo confidence bands and variance for the own-price coefficient $\hat{\gamma}_{11}$ across 100 replications.

Figure 11. Independent AR(1) prices



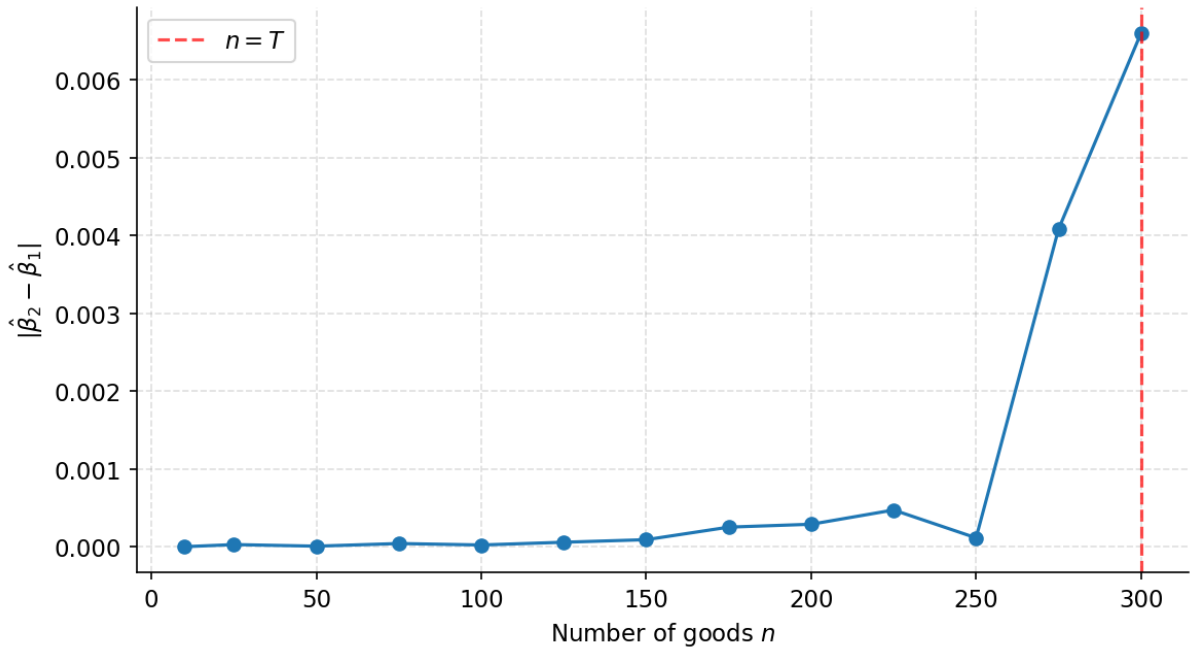
Notes: Simulation with $T = 300$ and independent AR(1) log-prices across goods. Without correlation across goods, the condition number rises modestly and OLS RMSE remains low up to the rank-deficient boundary.

Figure 12. Orthogonal regressors



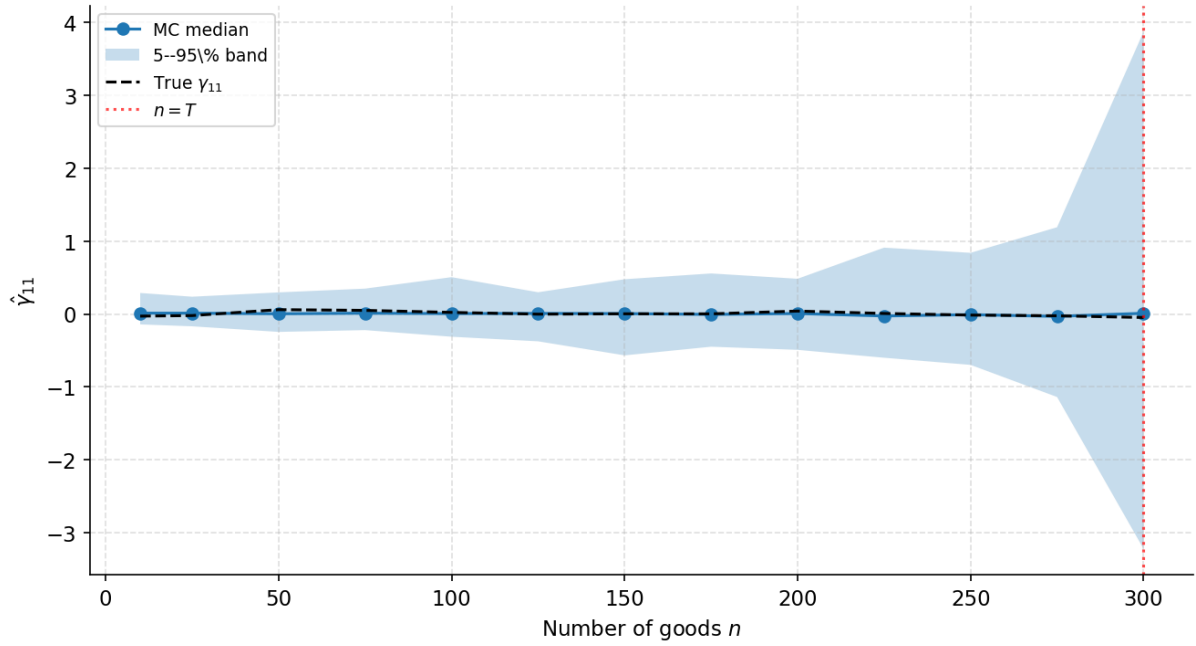
Notes: Simulation with $T = 300$ and orthogonal regressors, so that $X'X = TI_k$ exactly (perfect conditioning). OLS performance is unaffected by dimension until the rank-deficient boundary at $n_{\text{goods}} + 2 \geq T$, where RMSE rises sharply.

Figure 13. Perturbation sensitivity



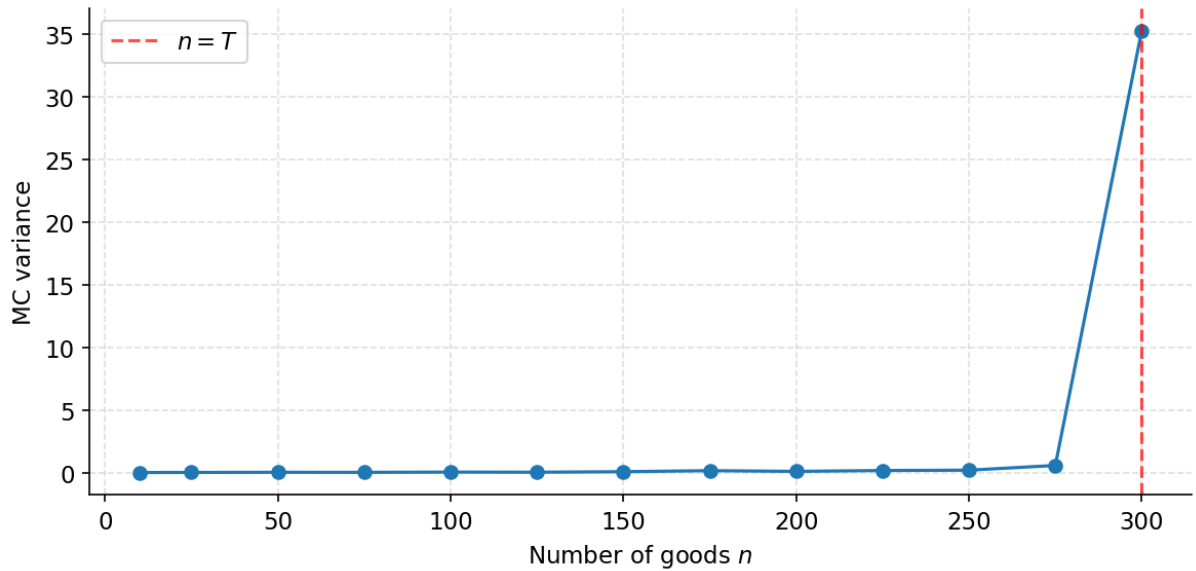
Notes: Sensitivity of OLS estimates $\hat{\beta}$ to small data perturbations under correlated prices. The plot shows $\|\hat{\beta}_2 - \hat{\beta}_1\|$, the change in the OLS coefficient vector when noise of magnitude $\varepsilon = 10^{-6}$ is added to X . Sensitivity stays low through moderate n_{goods} then spikes near the rank-deficient boundary, confirming the ill-conditioning documented in Figure 1.

Figure 14. Monte Carlo confidence bands for $\hat{\gamma}_{11}$



Notes: Monte Carlo confidence bands for the own-price coefficient $\hat{\gamma}_{11}$ across 100 replications under correlated prices. The 5–95% band stays tight through moderate dimensions then explodes near $n_{\text{goods}} = T$, reflecting the precision collapse documented in Figure 1.

Figure 15. Monte Carlo variance of $\hat{\gamma}_{11}$



Notes: Monte Carlo variance of $\hat{\gamma}_{11}$ across 100 replications under correlated prices. Variance stays low through moderate n_{goods} then rises sharply as n_{goods} approaches T , mirroring the precision collapse shown in Figure 14.

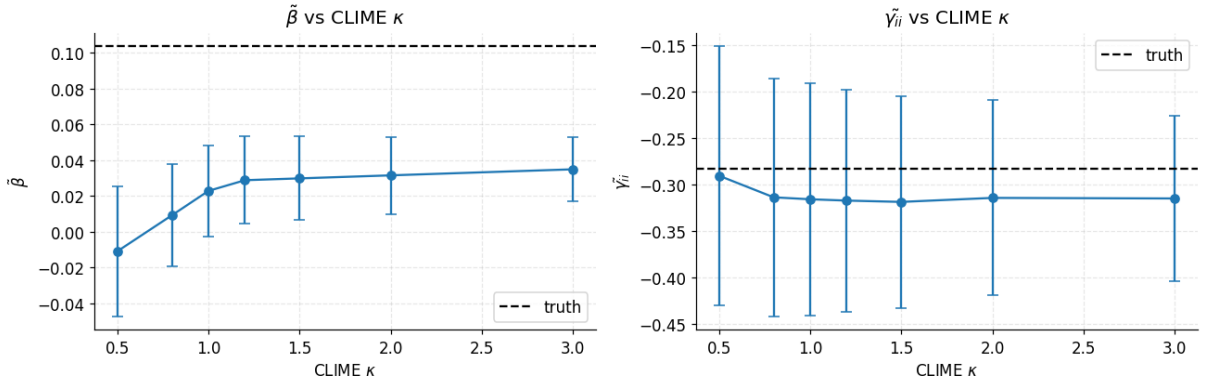
B Robustness checks

This appendix collects the robustness checks referenced in Section 5. Section B.1 verifies that the substantive findings are not driven by the CLIME tuning choice. Section B.2 completes the calibration check of Scenario A. Section B.3 reports the weak-instruments stress test of Scenario C. Section B.4 reports estimator-level robustness for Scenario B, varying the first-stage choice and the precision-matrix construction, and reports the precision-matrix variants that extend the estimator through the boundary $p_x \geq T$ where mainline CLIME is not computed.

B.1 Robustness to CLIME tuning

The mainline CLIME tuning is $\mu = 1.2\sqrt{\log p_x/T}$ following Gold et al. (2020). Figure 16 evaluates sensitivity to the tuning level κ over $[0.5, 3.0]$.

Figure 16. Robustness to CLIME tuning κ



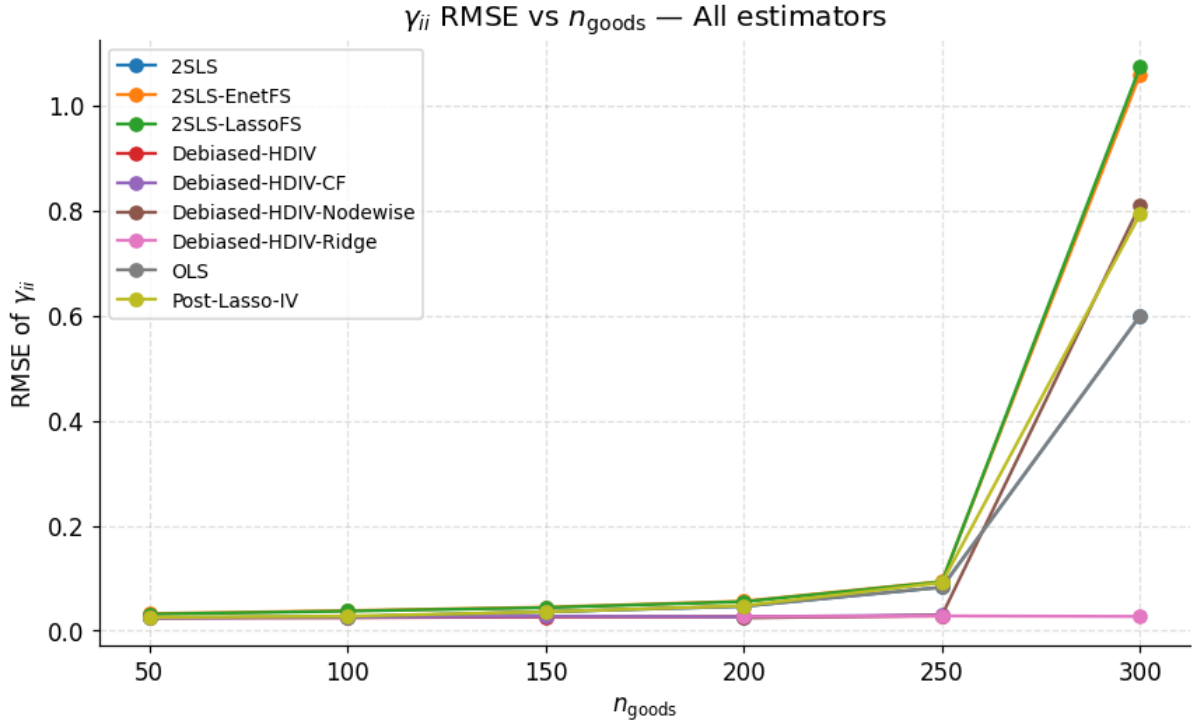
Notes: Debiased HD-IV with cross-fitting at a single replication in Scenario B ($n_{\text{goods}} = 100$, $T = 300$, $p_z = 350$). Each panel plots $\tilde{\beta}$ (left) and $\tilde{\gamma}_{ii}$ (right) at each $\kappa \in \{0.5, 0.8, 1.0, 1.2, 1.5, 2.0, 3.0\}$, with error bars giving sandwich 95% confidence intervals and the dashed line marking the true parameter. The paper uses $\kappa = 1.2$ throughout.

The $\tilde{\gamma}_{ii}$ interval contains the truth at every κ value while the $\tilde{\beta}$ interval misses it at every κ value. Both patterns are stable across the range, indicating that the substantive results are not driven by the tuning choice.

B.2 Scenario A

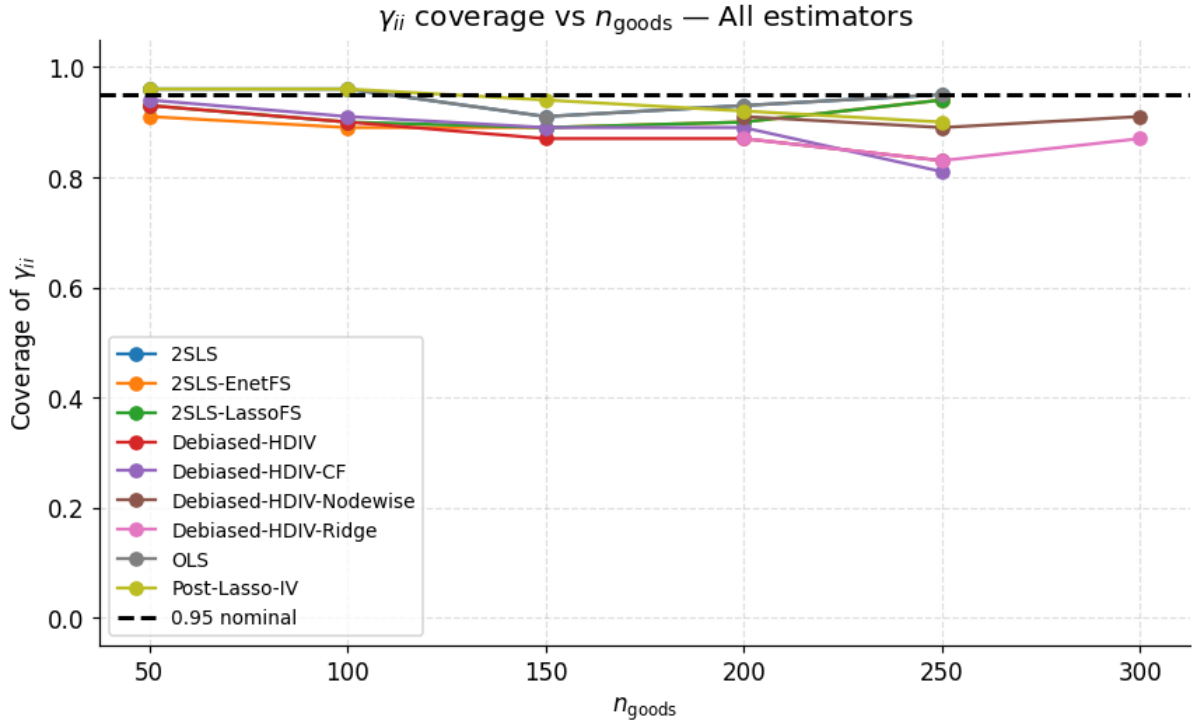
Scenario A sets $\rho_u = 0$, so prices are exogenous. Figures 17, 18, and 19 report the full results across the nine estimators.

Figure 17. Scenario A: RMSE of γ_{ii}



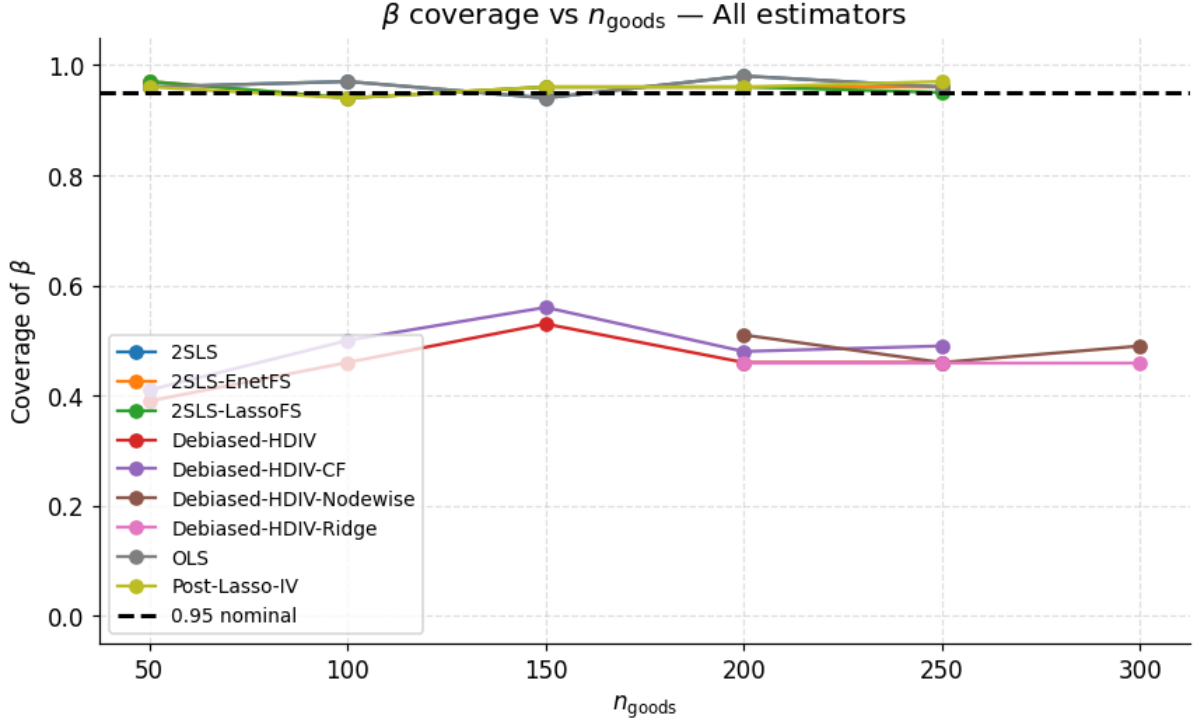
Notes: RMSE of γ_{ii} across the n_{goods} -grid at $T = 300$, $p_z = 350$, Scenario A ($\rho_u = 0$), $R = 100$. All nine estimators are plotted (including the three robustness variants).

Figure 18. Scenario A: coverage of γ_{ii}



Notes: Empirical coverage of nominal 95% confidence intervals for γ_{ii} across the n_{goods} -grid in Scenario A. Dashed line marks the nominal level. Several estimators are missing at the rank-deficient boundary.

Figure 19. Scenario A: coverage of β



Notes: Empirical coverage of nominal 95% confidence intervals for β across the n_{goods} -grid in Scenario A.

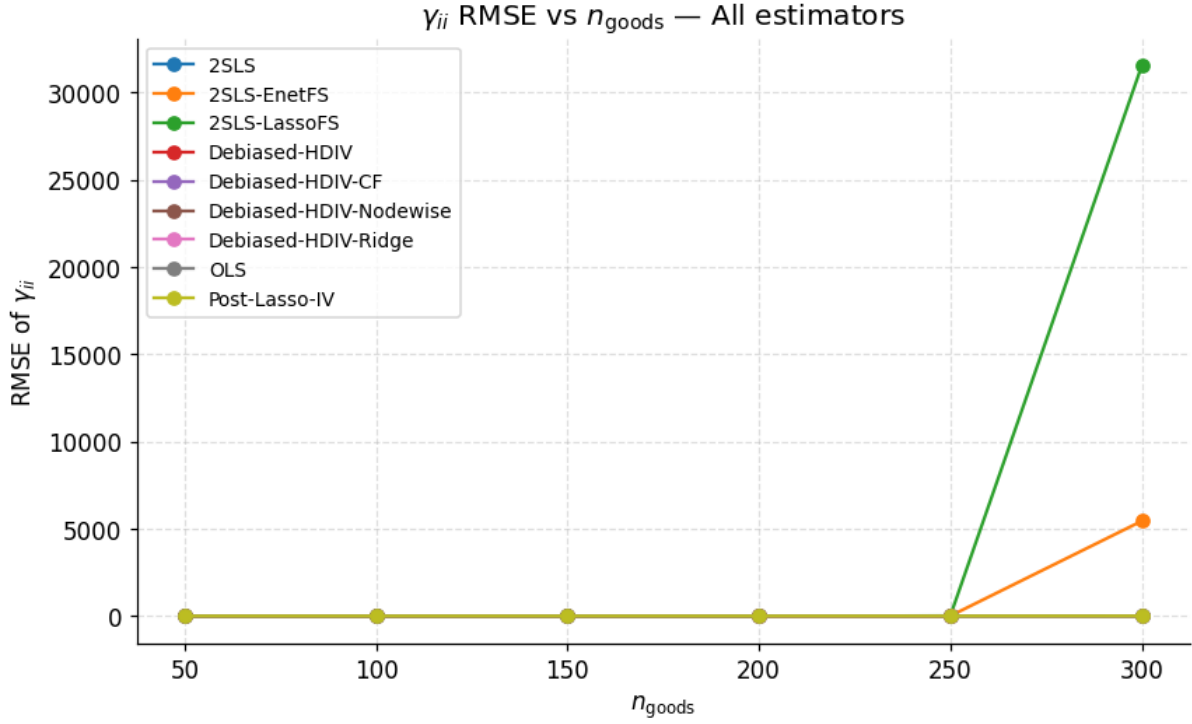
The RMSE picture matches expectations. OLS and classical 2SLS deteriorate as n_{goods} approaches T . At the boundary, only the ridge-regularized debiased variant maintains the RMSE level it held below, consistent with the discussion in Section 5.5. The β -coverage gap appears even without endogeneity, confirming the small-coefficient diagnosis.

B.3 Scenario C: weak instruments

Scenario C sets $\rho_u = 0.6$ with $k = 0.3$, giving weak identification. Under weak instruments, the IV-based estimators that rely on the first stage (2SLS, 2SLS-LassoFS, 2SLS-EnetFS, Post-Lasso IV, and the debiased variants) all face a noisier first stage; OLS does not use the instruments and is unaffected by their weakness, though it remains biased by the endogeneity. 2SLS-LassoFS and 2SLS-EnetFS spike by several orders of magnitude in RMSE. Post-Lasso IV is stable on RMSE but severely undercovers across p_z (the OLS-refit standard errors do not reflect the weak-instrument noise). The debiased variants remain more stable on RMSE than the Lasso first-stage variants but do not overcome

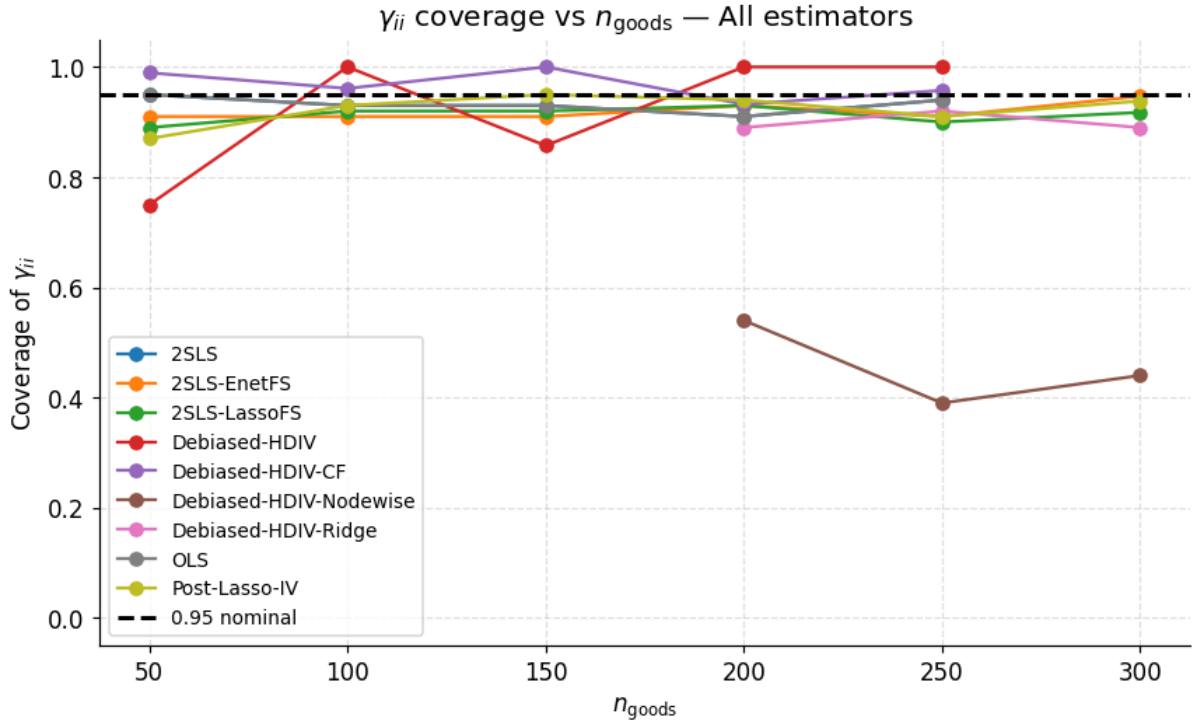
the identification failure, and the nodewise-Lasso variant develops its own coverage gap at higher n_{goods} .

Figure 20. Scenario C: RMSE of γ_{ii} in the n_{goods} -sweep



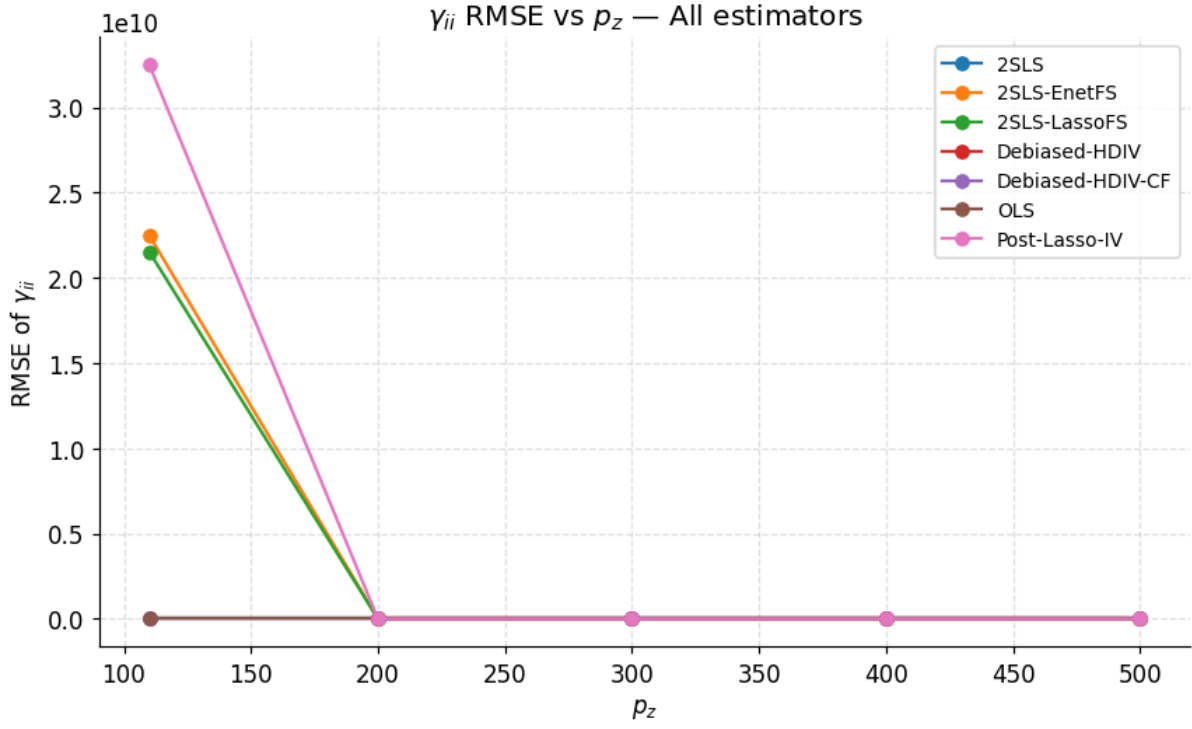
Notes: RMSE of γ_{ii} across the n_{goods} -grid at $T = 300$, $p_z = 350$, Scenario C, $R = 100$.

Figure 21. Scenario C: coverage of γ_{ii} in the n_{goods} -sweep



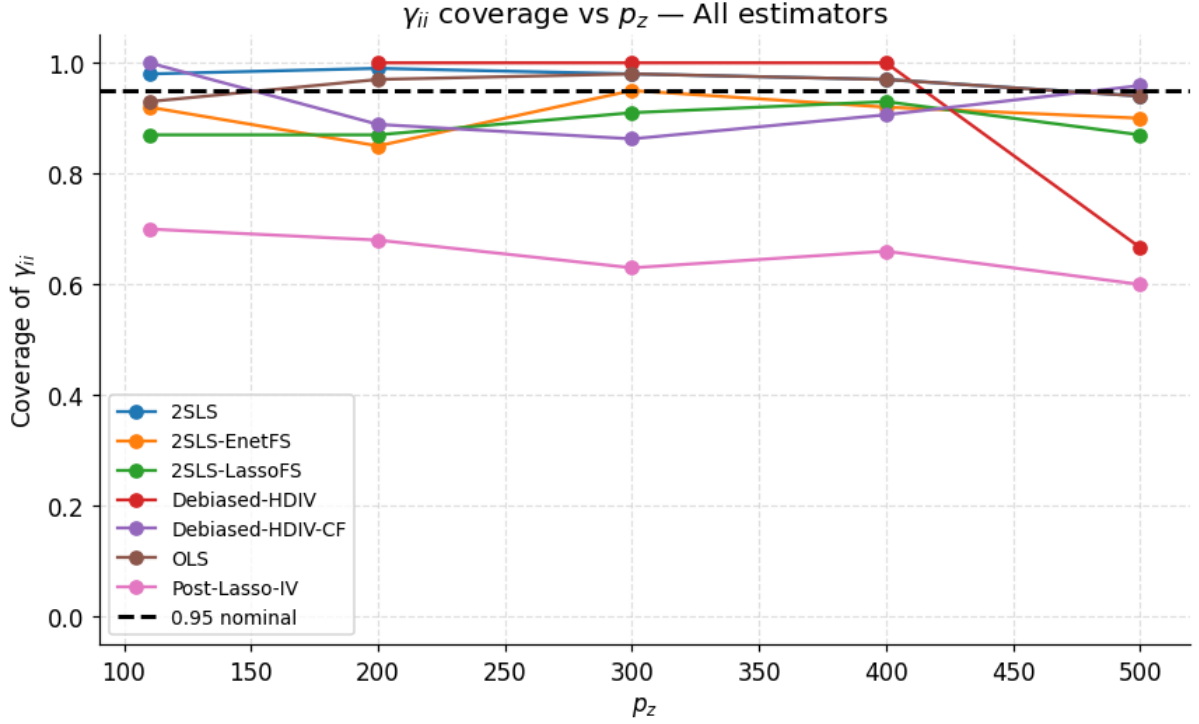
Notes: Empirical coverage of nominal 95% confidence intervals for γ_{ii} across the n_{goods} -grid in Scenario C. Dashed line marks the nominal level.

Figure 22. Scenario C: RMSE of γ_{ii} in the p_z -sweep



Notes: RMSE of γ_{ii} across $p_z \in \{110, 200, 300, 400, 500\}$ at fixed $n_{\text{goods}} = 100$, $T = 300$, Scenario C, $R = 100$.

Figure 23. Scenario C: coverage of γ_{ii} in the p_z -sweep

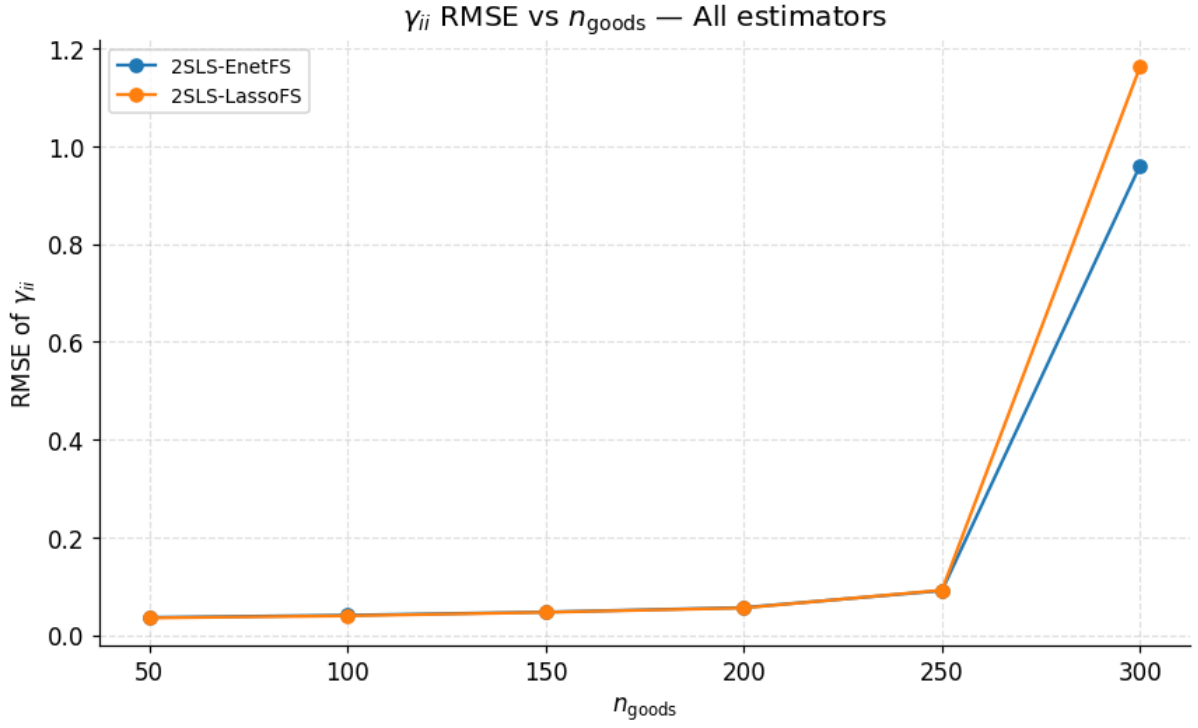


Notes: Empirical coverage of nominal 95% confidence intervals for γ_{ii} in the p_z -sweep, Scenario C. Dashed line marks the nominal level.

B.4 Estimator-level robustness within Scenario B

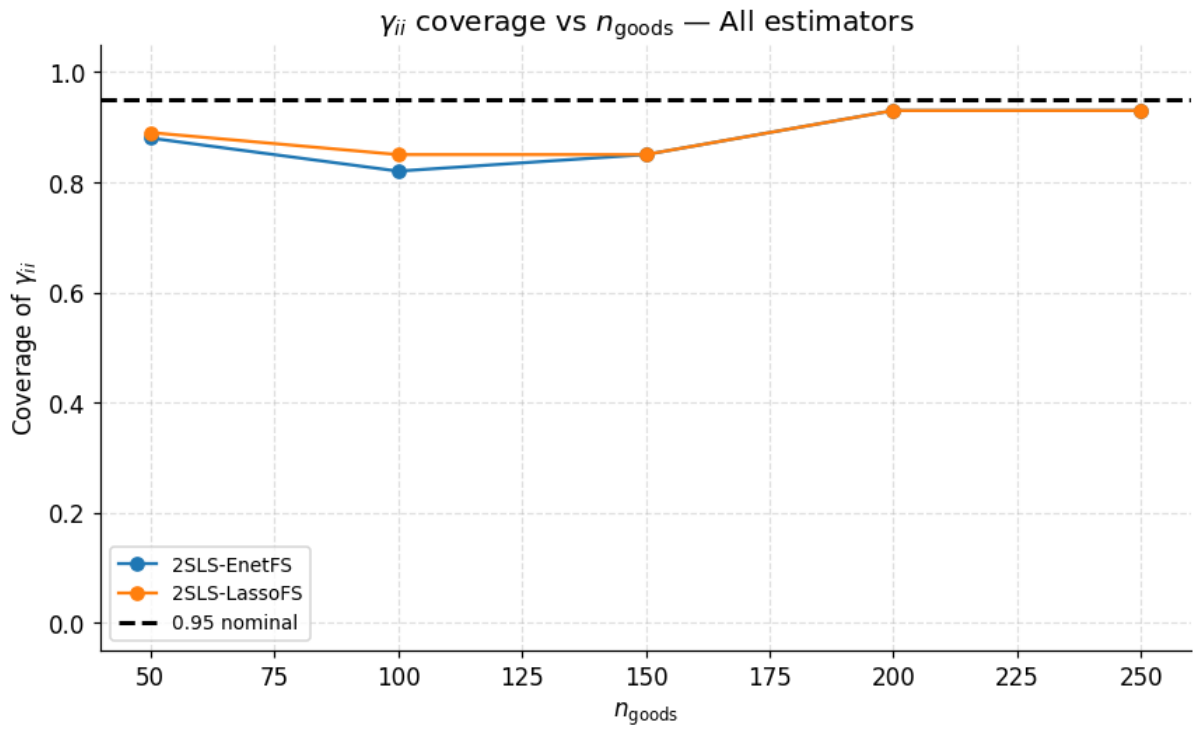
Two checks within Scenario B isolate whether the substantive conclusions depend on the specific first-stage or precision-matrix choice. The first-stage check compares 2SLS-LassoFS with the elastic-net first-stage variant (2SLS-EnetFS) of Bae (2018). The precision-matrix check compares the four debiased variants (mainline CLIME, cross-fitted CLIME, ridge-regularized, and nodewise Lasso) on γ_{ii} across the grid, and on β through the boundary $p_x \geq T$ where mainline CLIME is not computed.

Figure 24. First-stage robustness: RMSE of γ_{ii} , Scenario B



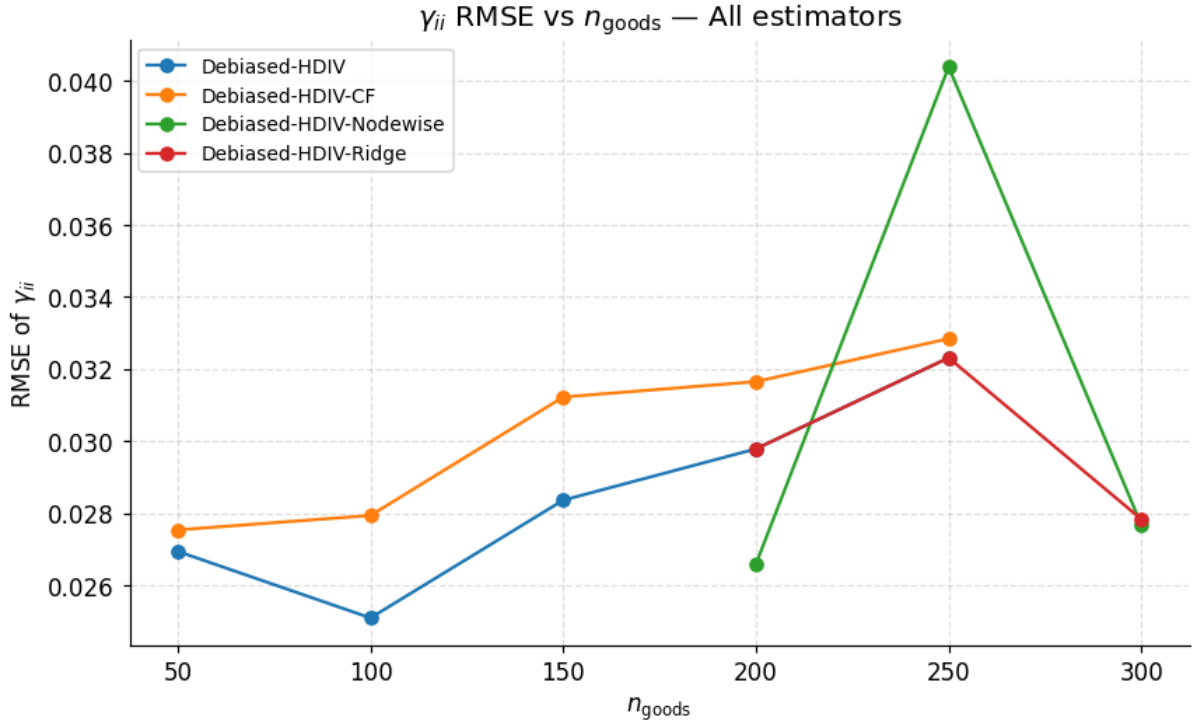
Notes: 2SLS-LassoFS versus 2SLS-EnetFS in Scenario B, $T = 300$, $p_z = 350$, $R = 100$.

Figure 25. First-stage robustness: coverage of γ_{ii} , Scenario B



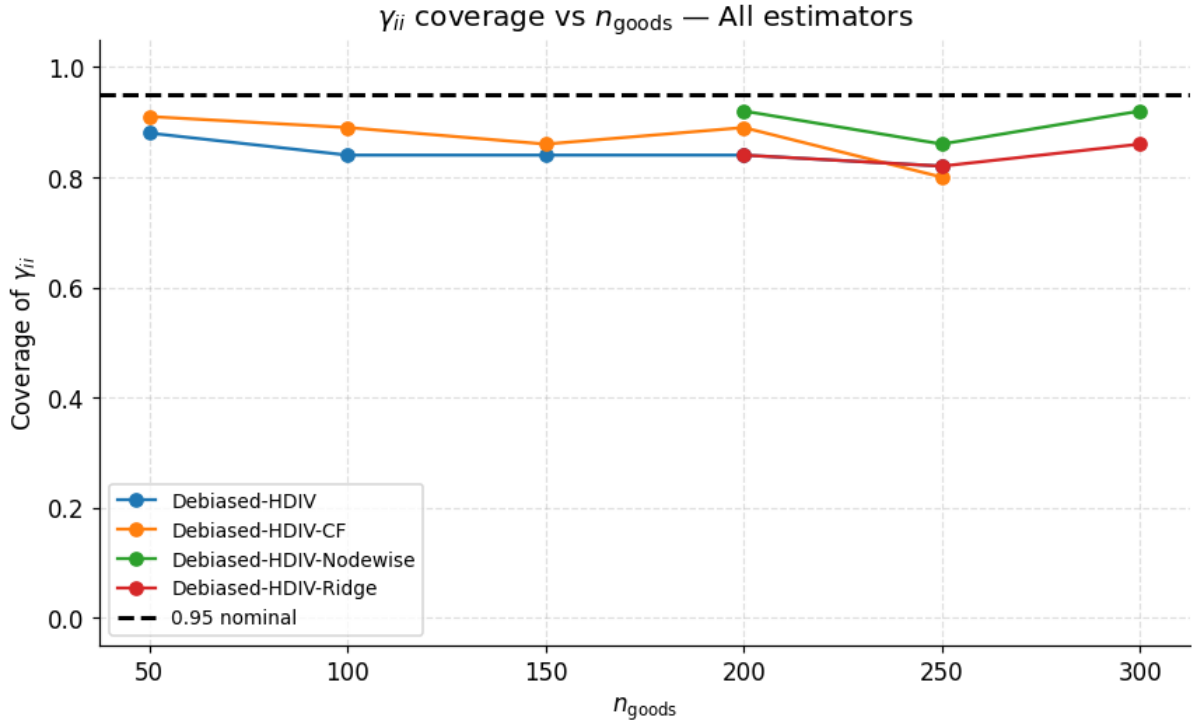
Notes: Coverage of nominal 95% confidence intervals for γ_{ii} comparing 2SLS-LassoFS and 2SLS-EnetFS, Scenario B. Dashed line marks the nominal level.

Figure 26. Precision-matrix robustness: RMSE of γ_{ii} , Scenario B



Notes: Debiased HD-IV variants compared in Scenario B, $T = 300$, $p_z = 350$, $R = 100$.

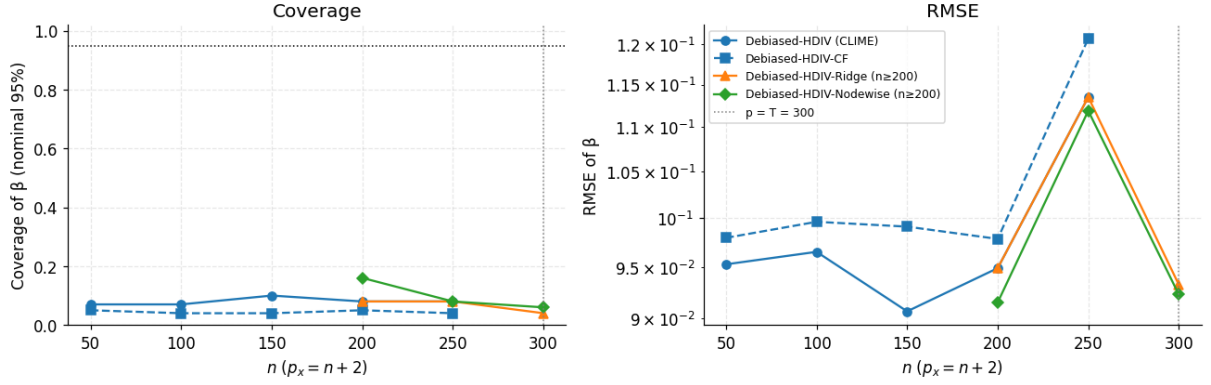
Figure 27. Precision-matrix robustness: coverage of γ_{ii} , Scenario B



Notes: Coverage of nominal 95% confidence intervals for γ_{ii} across the four debiased variants in Scenario B. Dashed line marks the nominal level.

The substantive conclusions of Section 5 are robust to the specific first-stage or precision-matrix choice. RMSE patterns are similar across the four debiased variants in the inner grid: all four sit in a narrow band, with no variant diverging from the others. At the boundary, the ridge and nodewise variants extend through with RMSE values comparable to what the mainline achieves below the boundary. All variants undercover for γ_{ii} at moderate levels below the nominal 0.95, consistent with the coverage pattern reported for the mainline in Section 5.5.1.

Figure 28. Boundary regime: extended debiased variants on β



Notes: Coverage (left) and RMSE (right) of β across $n_{\text{goods}} \in \{50, 100, 150, 200, 250, 300\}$ at $T = 300$, $p_z = 350$, Scenario B, $R = 100$. The ridge and nodewise-Lasso variants are plotted from $n_{\text{goods}} \geq 200$ and become the relevant debiased variants at the boundary $n_{\text{goods}} = 300$.

Figure 28 reports the four debiased variants on β across the grid and through the boundary. The ridge and nodewise variants recover the inner-grid picture and hold a narrow RMSE band, and the β -coverage gap from Section 5.5.1 persists across all four.

C Code and replication

All code to reproduce the simulations, figures, and tables in this paper is available at <https://github.com/sahmed95/Almost-Ideal-Demand-System-HD-IV/>.

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